

The rate of reaction

Learning outcomes

By the end of this section you should be able to:

- Determine rates of reaction from the tangents of graphs of concentration, volume or mass against time.
- Explain how changes in reaction rate can be monitored indirectly by monitoring changes in mass, volume or colour.

In the study of chemical reactions we are mainly concerned with the rate of reaction. Different reactions proceed at different rates. How can we tell whether a reaction is fast or slow? Generally we measure how fast the reactants disappear or how fast products form, over a given period of time. Fortunately, we do not have to try to count the actual number of molecules of the reactants or products. We can measure a quantity that is directly related to the number of molecules or their concentration. For example, for certain reactions where gases are produced the volume of gas produced over a given time can be measured easily and gives us the rate of the reaction.

The rate of reaction is expressed as the change in the concentration of the product or reactant per unit time.

$$\text{rate} = \frac{\text{change in concentration of reactants or products}}{\text{time}}$$

We know that as a reaction proceeds, reactant concentration decreases and product concentration increases. Watch the two parts of the same video in **Interactive 1** (left video first and use the full screen function if required) that show the change in concentration of reactants and products in a chemical reaction.

Interactive 1. Change in concentration of reactants and products in a chemical reaction.

 More information for interactive 1

1
00:00:00,240 --> 00:00:03,040
narrator: About chemical kinetics.
2
00:00:04,360 --> 00:00:09,640
Chemical kinetics is the study of rates
of chemical processes.
3
00:00:10,320 --> 00:00:13,320

The rate of reaction or speed of reaction

4

00:00:13,400 --> 00:00:18,080

can be defined as the change
in concentration of reactant or product

5

00:00:18,280 --> 00:00:20,560

in a particular reaction.

6

00:00:20,880 --> 00:00:23,080

Let us consider this

7

00:00:23,000 --> 00:00:27,080

round bottomed flask contains

N number of A molecules

8

00:00:29,320 --> 00:00:32,040

within a particular period of time.

9

00:00:33,080 --> 00:00:37,960

Like, here, T one is the initial time.

10

00:00:40,800 --> 00:00:42,720

After starting the timer,

11

00:00:43,560 --> 00:00:50,200

N number of A molecules

are converted into B molecules.

12

00:00:51,120 --> 00:00:56,920

Now we can see the disappearing
of A molecule that is reactant molecule,

13

00:00:57,000 --> 00:01:01,520

and appearing of B molecules,

that is product molecules.

14

00:01:04,080 --> 00:01:09,200

At the final time, T₂,

all the B molecules are formed

15

00:01:10,040 --> 00:01:14,160

and settle down

at the bottom of the RB flask.

16

00:01:14,240 --> 00:01:16,080

That is a round bottomed flask.

17

00:01:18,040 --> 00:01:22,080

Now we can write this reaction

as a chemical equation

18

00:01:25,400 --> 00:01:31,080

A is disappearing and B is appearing.

19

00:01:32,120 --> 00:01:34,880

Now we can write the rate.

20

00:01:35,400 --> 00:01:40,520

Rate is equal to change in concentration
divided by time taken.

21

00:01:43,080 --> 00:01:46,680

Mathematically it can be written as

22

00:01:48,320 --> 00:01:50,520

negative sign of ΔE .

23

00:01:50,840 --> 00:01:53,600

ΔE is the change in concentration.

24

00:01:54,080 --> 00:01:57,920

Concentration is always represented
within the square bracket

25

00:01:58,000 --> 00:01:59,720

divided by ΔT .

26

00:01:59,840 --> 00:02:02,600

Now we write the full form of ΔE

27
00:02:05,120 --> 00:02:07,720
concentration of A at T2 time

28
00:02:07,800 --> 00:02:13,120
minus concentration of A at T1 time
whole divided by T2 minus T1.

29
00:02:14,080 --> 00:02:17,320
We can write this rate of

30
00:02:19,560 --> 00:02:23,480
reaction in the differential format

31
00:02:24,000 --> 00:02:27,240
R is equal to minus D of A divided by DT

32
00:02:27,320 --> 00:02:31,200
or otherwise plus D of B divided by DT.

33
00:02:32,080 --> 00:02:33,720
Here, a negative sign indicates

34
00:02:33,800 --> 00:02:38,640
a decreasing in concentration
of reactant molecule .

35
00:02:38,720 --> 00:02:44,760
Positive sign indicates increasing
the concentration of product molecule.

36
00:02:45,320 --> 00:02:49,200
Now we draw a graph
between concentration versus time.

37
00:02:50,120 --> 00:02:56,440
In this graph, we can see
how the concentration of A

38
00:02:56,520 --> 00:03:03,040
decreases and the concentration
of B increases with the increasing time.

39
00:03:04,040 --> 00:03:06,000
In the next video, we will discuss

40
00:03:06,080 --> 00:03:11,240
how to write the rate of the reaction
for the stoichiometric equation.

41
00:03:11,320 --> 00:03:12,120
Thank you.

If we were to measure the rate of reaction as a change in the concentration of the products we could write the rate as shown in **Table 1**.

Table 1. Equation for rate as a change in the concentration of the products.

Equation	Symbols	Units
$\text{rate} = \frac{\Delta[P]}{\Delta t}$	rate	moles per cubic decimetre per second ($\text{mol dm}^{-3} \text{ s}^{-1}$)
	$\Delta[P]$ = the change in concentration of the products	moles per cubic decimetre (mol dm^{-3})
	Δt = the change in time	seconds (s)

Recall from section S1.4.5a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-concentration-of-a-solution-id-44263/>) that the square brackets mean concentration.

Similarly, if we were to measure the change in concentration of reactants the equation can be written as shown in **Table 2**.

Table 2. Equation for rate as a change in the concentration of the reactants.

Equation	Symbols	Units
$\text{rate} = -\frac{\Delta[R]}{\Delta t}$	rate	moles per cubic decimetre per second ($\text{mol dm}^{-3} \text{s}^{-1}$)
	$\Delta[R]$ = the change in concentration of the reactants	moles per cubic decimetre (mol dm^{-3})
	Δt = the change in time	seconds (s)

Study skills

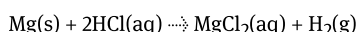
The negative sign in the rate equation in Table 2 indicates that the concentration of the reactants is decreasing but by convention, the rate of a reaction is always taken as a positive value.

Measuring rates of reaction

As you know, the rate of a reaction can be determined experimentally by measuring the change in concentration of the reactants or products. Whether we choose to measure the change in concentration of products or reactants depends on the nature of the reaction. Here are some of the techniques that can be used.

Measuring the change in volume of a gas

The reaction rate for the reaction between an acid and a metal to produce hydrogen gas can be found by measuring the volume of hydrogen produced over regular intervals. This will enable us to draw a graph of volume vs. time.



Gas collection can be done in several ways. A gas syringe can be used to collect the gas. The gas syringe is specially designed with a glass barrel that moves outward as it fills with gas. As the syringe is calibrated, the volume of gas can be measured directly and at the same time a stopwatch can be used to measure the time (**Figure 1**).

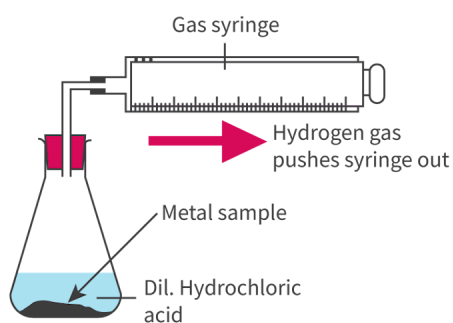


Figure 1. Collection of hydrogen gas using the gas syringe.

 More information for figure 1

The diagram illustrates the process of collecting hydrogen gas using a gas syringe. It features a setup with a flask that contains a metal sample and diluted hydrochloric acid. A line connects the flask to a gas syringe, which is shown extending as it fills with hydrogen gas. An arrow points from the hydrogen gas, labeled as pushing the syringe outward. The design shows how the gas syringe is calibrated to measure the volume of hydrogen gas collected during the reaction, where the metal interacts with the acid to produce the gas.

[Generated by AI]

Alternatively, the data logging equipment shown in **Figure 2** can be used to measure the quantity of gas produced by using a gas pressure sensor. The volume of gas produced is directly proportional to the pressure of the gas (**Figure 3**). Pressure changes are particularly useful for reactions in which both reactants and products are gaseous.

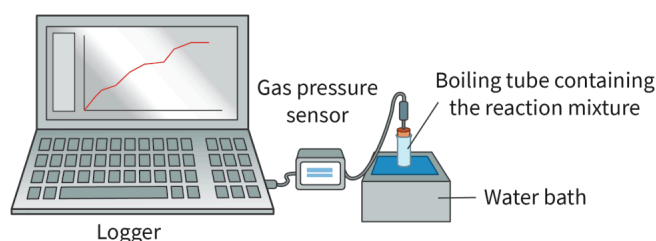


Figure 2. A gas pressure data logger.

 More information for figure 2

The image is an illustration showing a gas pressure sensor setup connected to a laptop acting as a data logger. The system includes a boiling tube containing the reaction mixture submerged in a water bath. A gas pressure sensor is connected to the boiling tube, which is in turn connected to the laptop. The screen of the laptop displays a red line graph representing pressure data over time, as the data logger records the changes. Labels in the image identify each component: "Logger" next to the laptop, "Gas pressure sensor" near the sensor apparatus, "Boiling tube containing the reaction mixture" pointing to the tube, and "Water bath" at the base where the tube is situated.

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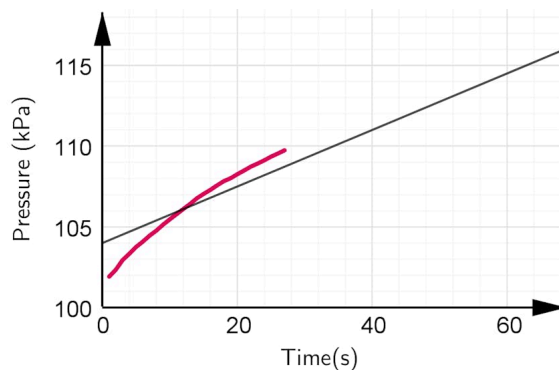


Figure 3. Results of an experiment to determine the rate of reaction by following change in pressure against time using a gas pressure sensor.

 More information for figure 3

The graph displays pressure in kilopascals (kPa) on the Y-axis against time in seconds (s) on the X-axis. The Y-axis ranges from 100 to 115 kPa, and the X-axis ranges from 0 to 60 seconds. There are two lines present: one is a red line depicting experimental

data, which starts slightly above 100 kPa and rises to just above 110 kPa by approximately 30 seconds, showing a non-linear increase initially and leveling out somewhat. The other line is a straight, black line showing a steady increase, serving as a reference or control.

[Generated by AI]

Measuring the changes in mass

Many reactions in which gases are produced will show a decrease in mass if the gas is removed from the reaction mixture. This change in mass can be measured over time and the rate of reaction determined. The rate of a reaction between a metal and acid producing hydrogen gas can be monitored with this method, as shown in **Figure 4**.

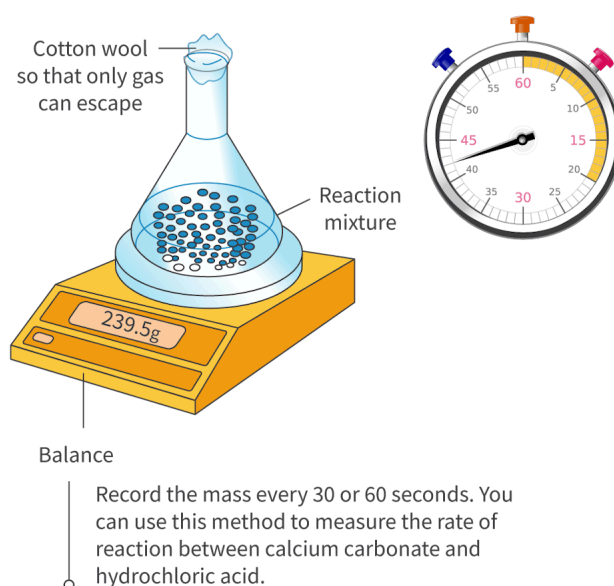


Figure 4. Measuring the rate of reaction by following changes in mass against time.

More information for figure 4

The illustration shows a laboratory setup to measure the rate of reaction through changes in mass over time. A conical flask is placed on an electronic balance that reads 239.5g. The flask contains a reaction mixture represented by small granular particles. It is covered with cotton wool, described as allowing only gas to escape. Next to the balance, there is a large stopwatch displaying time intervals with notable markings at every 5-second increment, highlighting the 30-, 45-, and 60-second points. The text suggests recording the mass every 30 or 60 seconds and notes that this method is used for reactions, such as those between calcium carbonate and hydrochloric acid.

[Generated by AI]

Measuring the changes in colour

A colorimeter or spectrophotometer is a device that measures the amount of light transmitted or absorbed by a coloured solution. Light of specific wavelength is selected and passed through the solution sample and the absorbance (or transmittance) is measured and plotted against time. We know that coloured solutions absorb some parts of the visible light spectrum and transmit others. As the concentration of some solutions increases, they become darker and absorb proportionally more light. This means less light is transmitted.

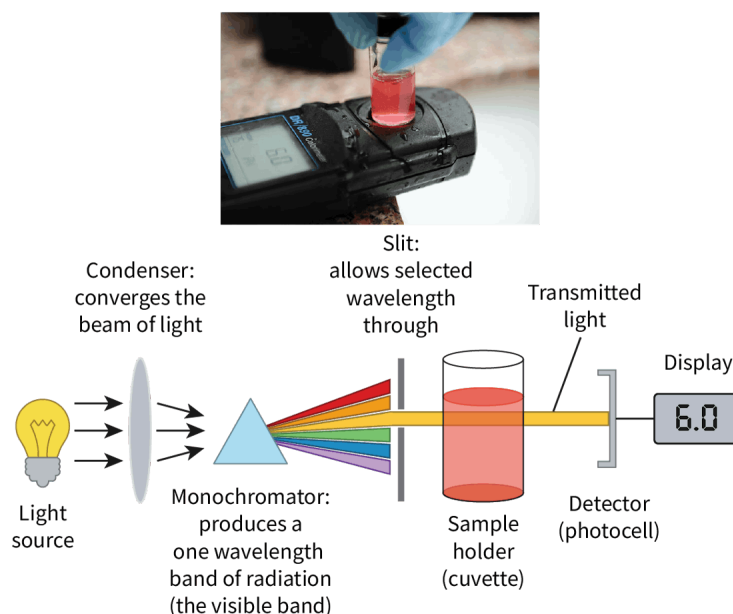


Figure 5. The simplified working of a colorimeter.

 More information for figure 5

The image is a diagram illustrating the working components of a colorimeter accompanied by an inset photo of an actual colorimeter in use. It starts with a bulb labeled as 'Light source,' which emits light. This light passes through a condenser labeled 'Condenser: converges the beam of light,' forming a focused beam of light. The beam then enters a monochromator, depicted as a prism, labeled 'Monochromator: produces a one wavelength band of radiation (the visible band).' The monochromator releases a spectrum from which light of a specific wavelength is selected and directed through a slit with the label 'Slit: allows selected wavelength through.' The light then travels through a sample holder labeled 'Sample holder (cuvette),' filled with a colored solution. Next, the light labeled 'Transmitted light' travels to a detector, also referred to as 'photocell,' which measures the light intensity. Finally, a digital output labeled 'Display' shows the measurement as '6.0.' The inset photo above shows the top view of a real colorimeter with a hand placing a sample into it.

[Generated by AI]

The reaction between propanone and iodine starts off with a yellow brownish colour of iodine and gradually fades to colourless. This means we can follow the reaction using colorimetry.



As the reaction moves towards completion and the solution becomes lighter in colour the absorbance values decrease.

Certain colorimeters can also be attached to data loggers to obtain continuous readings that are then represented in a graphical form. **Video 1** explains how to operate a simple colorimeter in the laboratory.

Using colorimeters



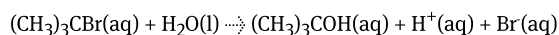
Video 1. Using a colorimeter.

Other methods for measuring reaction rates

Measuring changes in conductivity

A change in the concentration of ions in solution leads to a change in conductivity. Hence reactions, that involve changes in ion concentration when reactants change into products, can be monitored by taking conductivity measurements (**Figure 6**).

For example,



In this reaction, an increase in the concentration of ions in the solution (2 moles of ions on the right-hand side and 0 moles of ions on the left-hand side) will manifest as an increase in conductivity during the progress of the reaction.

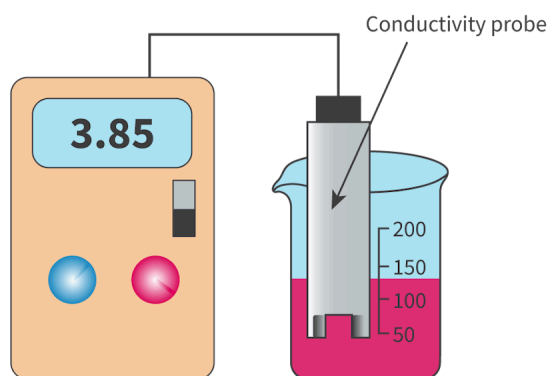


Figure 6. An electrical conductivity meter.

More information for figure 6

The image depicts an electrical conductivity meter connected to a probe. On the left, there is a digital device displaying the value 3.85. The device has two colored buttons below the display, one blue and one pink, and a gray switch on the right side. A wire leads from this device to a conductivity probe inserted into a transparent beaker filled with pink liquid. The beaker has measurement markings on the side, indicating levels at 50, 100, 150, and 200 units. The probe has an arrow pointing downward into the liquid, labeled 'Conductivity probe.' The image illustrates a setup for measuring the conductivity of a liquid, likely related to a chemical reaction where ion increase is measured.

[Generated by AI]

Measuring the rate of formation of precipitate

The sodium thiosulfate and hydrochloric acid reaction produces a pale yellow precipitate of sulfur over a period of time which results in the clear solution turning opaque.



The progress of this reaction can be followed by measuring the time taken for the solution to turn opaque, so that a cross viewed through the solution is no longer visible (**Figure 7**).

A video of the 'disappearing cross' reaction can be seen in the next [section R2.2.2-3a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/factors-affecting-the-rate-of-reaction-id-44714/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/factors-affecting-the-rate-of-reaction-id-44714/>), where the effect of concentration on the reaction is investigated.

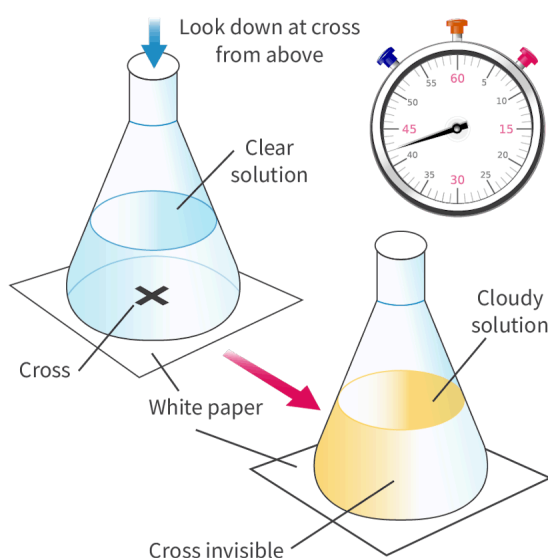


Figure 7. Recording the increase in opaqueness during a reaction.

 More information for figure 7

The image is a diagram illustrating the 'disappearing cross' experiment. It contains two conical flasks positioned over a cross drawn on paper. The first flask shows a clear solution, through which the cross is visible. An arrow indicates the progression from the first to the second flask, which contains a cloudy solution, obscuring the cross. A stopwatch is displayed beside the flasks, marking time intervals. Labels identify the 'clear solution', 'cloudy solution', and the 'cross invisible' condition as the reaction progresses.

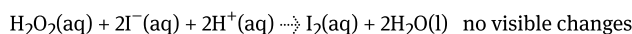
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Measuring the rate of clock reactions

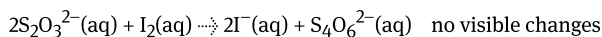
A clock reaction generally involves a series of chemical reactions where an additional reaction can only occur after the reaction(s) under study is/are complete. This clock reaction will involve a very rapid colour change or formation of a precipitate. By measuring the time taken for the visible change of the final reaction in the series to occur or repeat itself, chemists can measure the rate of the reaction under study.

For example, in the iodine clock reaction two colourless solutions are mixed, one containing hydrogen peroxide and sulfuric acid and the other containing potassium iodide, sodium thiosulfate and starch. At first there is no visible reaction but only when the first two reactions are completed, can the final reaction occur, and the mixture turns dark blue (**Figure 8**).

In the first set of reactions iodine is produced



This iodine is then used up by the thiosulfate.



The amounts chosen are such that as soon as the iodine is in excess (and the thiosulfate is limiting) the iodine reacts with the starch to form a blue-black complex.

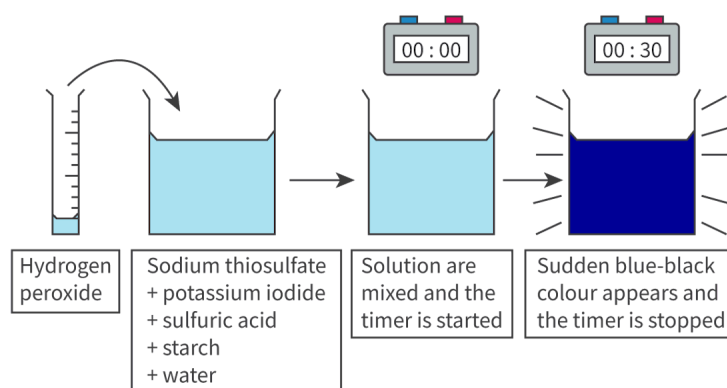
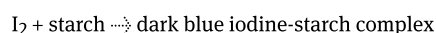


Figure 8. The iodine clock reaction.

 More information for figure 8

The image is a diagram depicting the iodine clock reaction process in four steps.

- 1. Hydrogen Peroxide Introduction:** A beaker is shown with a label "Hydrogen peroxide." This is added to another solution.
- 2. Solution Mixture:** A new beaker, now containing a blue liquid, is labeled with its components: sodium thiosulfate, potassium iodide, sulfuric acid, starch, and water.
- 3. Timer Started:** The mixture is stirred, initiating the reaction, marked by a digital timer displaying 00:00.
- 4. Color Change and Timer Stop:** The solution changes rapidly to a blue-black color, indicated by a visual cue from a different beaker, and the timer shows 00:30. Text states that the color change is sudden, signaling the reaction completion.

[Generated by AI]

Spot reactions where both reactants are aqueous, and a product is produced in the form of a precipitate. This is shown by the state symbol for a solid 's' next to the product, as in the production of sulfur in the sodium thiosulfate and hydrochloric acid reaction. A precipitate is always opaque and the rate of such reactions can be measured by monitoring changes in solution clarity.

Calculating reaction rates from graphs

The graph in **Figure 9** shows the change in concentration of products for two different reactions. We can see that products are formed more quickly in the blue curve as compared to the red curve. That is, the curve is steeper. Hence the steepness of the curve is an indication of 'how quickly' the reaction proceeds. The steepness of the curve can be measured mathematically by calculating the gradient or slope of the line. The gradient of a line is calculated by measuring the change in concentration over time and is therefore a measure of the rate of the reaction.

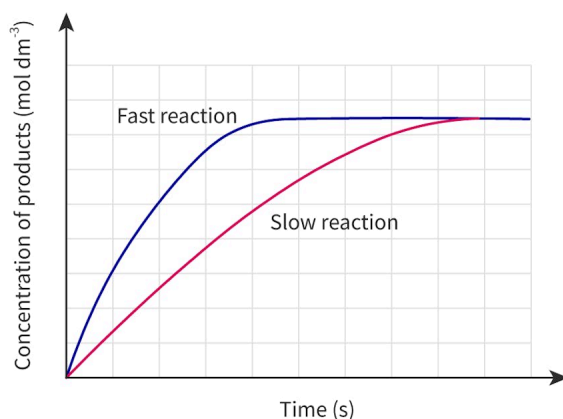


Figure 9. A graph comparing rates of reaction.

 More information for figure 9

The image is a graph displaying the rates of reactions by comparing two different reaction curves over time. The X-axis is labeled "Time (s)" and the Y-axis is labeled "Concentration of products (mol dm⁻³). There are two curves displayed on the graph: a blue curve and a red curve. The blue curve, labeled "Fast reaction," shows a steep increase in the concentration of products, indicating a rapid reaction rate. In contrast, the red curve, labeled "Slow reaction," shows a more gradual increase in concentration, representing a slower reaction rate. The steepness of each curve signifies the speed of the respective reaction. As time progresses, both curves plateau, indicating that the reactions have reached completion, with the fast reaction achieving this state quicker. The grid lines on the graph provide a visual reference for estimating values but are not numerically labeled.

[Generated by AI]

Calculating the instantaneous rate of a reaction

As reactions proceed, the rate decreases over time until one or more of the reactants is used up and the rate becomes zero. This means the rate of reaction or gradient keeps changing so instantaneous rate is often taken which is the rate at a particular time. This can be done by drawing a tangent to the graph at that particular time. **Figure 10** shows how the instantaneous rate can be measured at 12 seconds.

Calculating the initial rate of a reaction

The shape of a typical rate curve (**Figure 10**) shows that the rate of reaction is greatest at the start and slows down as the reaction proceeds. The initial rate measures the rate of reaction at the start of the reaction. Initial rates data is

important when comparing the effect of various factors on the reaction rates.

The initial rate of reaction can be calculated by drawing a tangent at 0 s and calculating the gradient as shown.

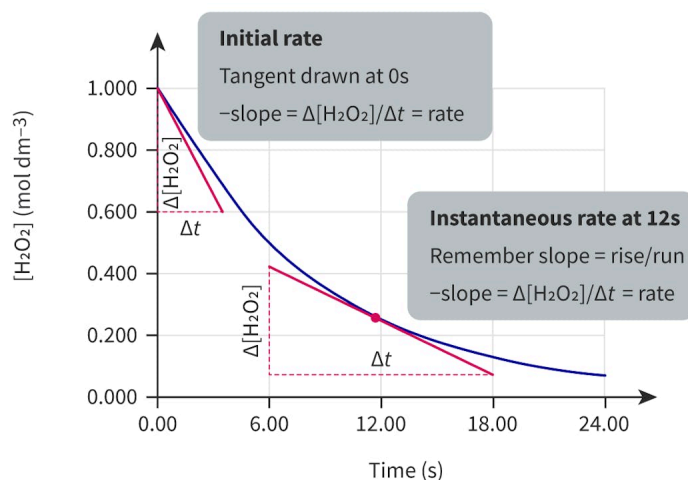


Figure 10. Graph of concentration vs. time for the decomposition of hydrogen peroxide. Tangents are shown at $t = 0$ s and $t = 12$ s.

[More information for figure 10](#)

The graph displays the concentration of hydrogen peroxide $[H_2O_2]$, measured in mol dm^{-3} , on the Y-axis versus time in seconds on the X-axis. The Y-axis ranges from 0.000 to 1.000 mol dm^{-3} , while the X-axis spans from 0 to 24 seconds. The curve represents the decrease in concentration over time, showcasing the decomposition process of hydrogen peroxide. Two tangents are drawn: one at time 0 seconds and another at time 12 seconds. The tangent at $t=0$ seconds intersects the graph at 1.000 mol dm^{-3} and corresponds with a slope given by the rate of reaction formula $\Delta[H_2O_2]/\Delta t$. Similarly, the tangent at $t=12$ seconds relates to the instantaneous rate of reaction at that time, emphasizing that the slope reflects the rate as $-\Delta[H_2O_2]/\Delta t$. The graph visually aids in understanding the initial and instantaneous rates of decomposition by illustrating the changes in slopes and concentration over time.

[Generated by AI]

Calculating the average rate of a reaction

At times when reactions proceed very quickly it is not easy to measure the instantaneous rate of the reaction and the average rate is taken. The average rate of reaction is an average of all the instantaneous reaction rates over a given period of time. The average rate of the reaction can be calculated by drawing a secant to the rate curve and measuring the gradient, as in **Figure 11**.

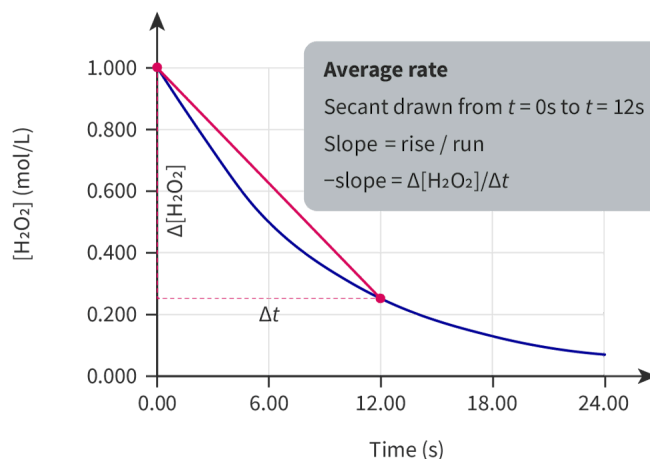


Figure 11. A graph of concentration vs. time for the decomposition of hydrogen peroxide. Secant shown from $t = 0\text{ s}$ to $t = 12\text{ s}$, to calculate the average rate in the first 12 s.

[More information for figure 11](#)

The image is a graph representing the concentration of hydrogen peroxide ($[\text{H}_2\text{O}_2]$) over time. The X-axis is labeled "Time (s)" and ranges from 0 to 24 seconds, while the Y-axis is labeled " $[\text{H}_2\text{O}_2]$ (mol dm^{-3})" ranging from 0.000 to 1.000. A curve plotted in blue shows the decomposition of hydrogen peroxide over time, descending from the top left (1.000 mol dm^{-3} at 0 seconds) towards the x-axis, indicating a decrease in concentration as time progresses. A red secant line is drawn from the point where time equals 0 to the point at 12 seconds to calculate the average rate of change in this interval. The gray box within the graph provides a description of the average rate, showing that the slope of the secant is calculated by the formula: Slope = rise/run = $\Delta[\text{H}_2\text{O}_2] / \Delta t$.

[Generated by AI]

Worked example 1

Figure 12 shows the volume of gas produced during a reaction.

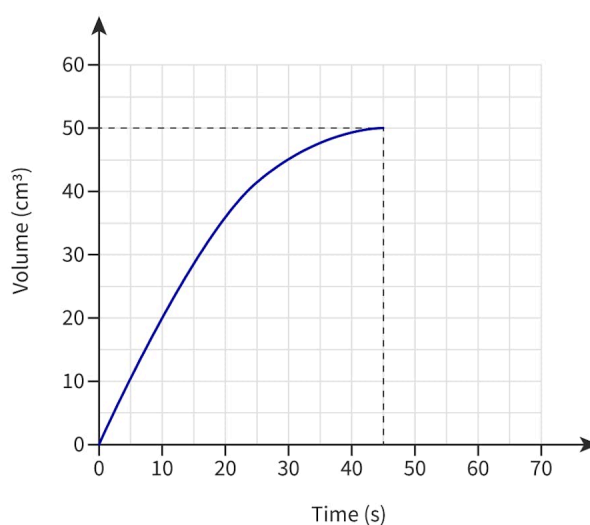


Figure 12. Volume vs. time graph for a reaction.

[More information for figure 12](#)

The image is a graph illustrating the relationship between volume and time for a chemical reaction. The X-axis represents time in seconds, ranging from 0 to 70 seconds, marked at intervals of 10 seconds. The Y-axis denotes volume in cubic

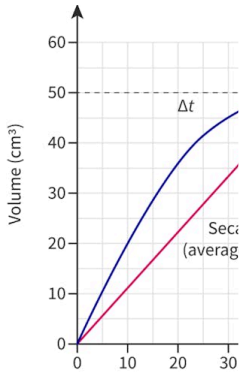
centimeters (cm^3), ranging from 0 to 60 cm^3 , with an interval of 10 cm^3 . A curve on the graph shows an increase in volume over time, starting at the origin (0,0) and rising sharply upwards to approach 50 cm^3 by the time it reaches approximately 50 seconds. The curve then levels out, showing a plateau as it reaches the maximum volume at around 50 cm^3 . Dashed lines extending horizontally and vertically from the curve to the axes indicate this plateau point.

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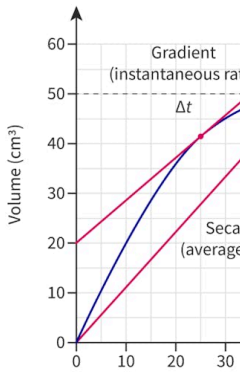
Find

- the average rate of the reaction from the graph,
- the instantaneous rate at 25 s.

- Calculating the average rate of reaction.

Solution steps	Calculations
Step 1: Draw the secant line.	
Step 2: Write out the required formula.	Find the gradient of the secant $\text{gradient} = \frac{\text{rise}}{\text{run}}$
Step 3: Write out the known values.	$\text{rise} = (50 - 0) = 50 \text{ cm}^3$ $\text{run} = (45 - 0) = 45 \text{ s}$
Step 4: Substitute the values given.	$= \frac{50}{45}$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$= 1.1 \text{ cm}^3 \text{ s}^{-1} \text{ (2 s.f.)}$ The unit for rate of reaction is $\text{cm}^3 \text{ s}^{-1}$. This is acceptable when the rate is measured as the volume of gas produced.

- Calculating the instantaneous rate of reaction at 25 s.

Solution steps	Calculations
Step 1: Draw the tangent line.	
Step 2: Write out the required formula.	Find gradient of tangent line ϵ $\text{gradient} = \frac{\text{rise}}{\text{run}}$
Step 3: Write out the known values.	$\text{rise} = (57-20) = 37 \text{ cm}^3$ $\text{run} (43-0) = 43 \text{ s}$
Step 4: Substitute the values given.	$= \frac{37}{43}$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$= 0.86 \text{ cm}^3 \text{ s}^{-1} (2 \text{ s.f.})$

Creativity, activity, service

Strand: Service

Learning outcomes:

- Demonstrate engagement with issues of global significance
- Recognise and consider the ethics of choices and actions

The average person produces approximately 2 kg of solid waste every day. While some of this waste can be recycled, about half of it ends up in [landfill sites](https://www.eastsussex.gov.uk/rubbish-recycling/how-we-manage-our-waste/landfill-sites/what-is-a-landfill-site#:~:text=A%20landfill%20site%20is%20an,unwanted%20hole%20in%20the%20ground.) [\[https://www.eastsussex.gov.uk/rubbish-recycling/how-we-manage-our-waste/landfill-sites/what-is-a-landfill-site#:~:text=A%20landfill%20site%20is%20an,unwanted%20hole%20in%20the%20ground.\]](https://www.eastsussex.gov.uk/rubbish-recycling/how-we-manage-our-waste/landfill-sites/what-is-a-landfill-site#:~:text=A%20landfill%20site%20is%20an,unwanted%20hole%20in%20the%20ground.).

Once in the landfill site, this waste decomposes but only very slowly depending on factors such as the amount of oxygen available, the temperature and the moisture. The decomposition of this waste can also produce [greenhouse gases](https://www.epa.gov/lmop/basic-information-about-landfill-gas) [\[https://www.epa.gov/lmop/basic-information-about-landfill-gas\]](https://www.epa.gov/lmop/basic-information-about-landfill-gas) such as methane, CH₄.

Think about what you can do to reduce the amount of waste produced by your school or in your local community — what can you do to reduce the amount of waste that ends up in landfill? Perhaps you could start by raising awareness of issues such as these as a possible CAS experience?

Activity

- **IB learner profile attribute:**
 - Knowledgeable
 - Inquires
 - Thinkers
 - Reflective
- **Approaches to learning:**
 - Thinking skills — Providing a reasoned argument to support a conclusion; Applying key ideas and facts in new contexts
 - Communication skills — Using terminologies, symbols and communication conventions consistently and correctly
 - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 25—30 minutes
- **Activity type:** Individual/pair activity

Work in pairs to complete the following tasks:

1. For the reaction between marble chips (calcium carbonate) and hydrochloric acid the rate of reaction can be measured in two ways, illustrated in **Figure 13**.

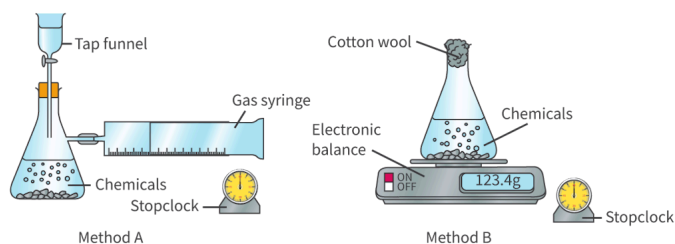


Figure 13. Two methods for measuring rate of reaction between marble chips and dilute hydrochloric acid.

 More information for figure 13

The image shows two different setups for measuring the reaction rate between marble chips and dilute hydrochloric acid. On the left, Method A involves a conical flask with chemicals labeled inside. A tap funnel is placed at the top of the flask for acid addition, and there's a gas syringe connected to the side to measure gas output. A stopclock is placed nearby to track the time of reaction. On the right, Method B uses a conical flask containing similar chemicals and has cotton wool plugged at the mouth to prevent spillage while allowing gas escape. This flask is placed on an electronic balance displaying 123.4g to measure mass change. A stopclock is also present to record the time. Each method aims to observe the rate at which carbon dioxide is produced when marble chips react with the acid.

[Generated by AI]

- (a) Sketch a graph of the carbon dioxide produced against time in Method A.
- (b) Explain why it is important to have a tap funnel that allows acid to be added when needed?
- (c) Sketch the graph you would obtain if you plotted the mass of the flask, shown by the electronic balance, against time in method B.
- (d) Sketch the graph you would expect if you plotted the mass of carbon dioxide given off against time in method B.
- (e) Explain why cotton wool is plugged into the mouth of the conical flask in Method B.

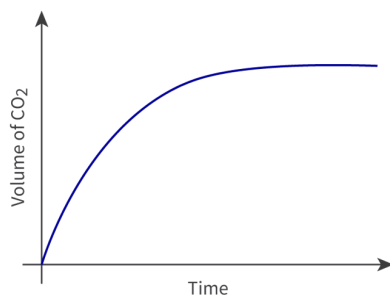
1. You are provided with a table containing six reactions. Five possible options of experimental techniques that may be used to measure the rate of each reaction, are given at the bottom of the sheet. Download the worksheet and complete the table by writing the correct experimental technique next to its respective reaction. More than one option can be

chosen for a reaction.

[Worksheet](#) 

(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry%20IB%20DP%20Chemistry%20SL%20HL%20FE2025%20Worksheet%201.pdf)

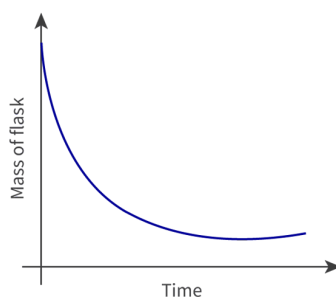
1. (a)



Ⓢ

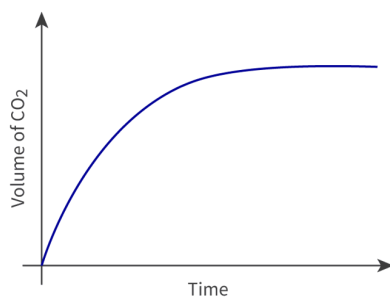
(b) The funnel allows hydrochloric acid to be added when the entire setup is sealed and ready to go. If we were to add the acid by opening the flask, this may result in the loss of gas as soon as it is produced, leading to an error in measuring the initial rate of the reaction.

(c)



Ⓢ

(d) Same as in part (a).



Ⓢ

(e) The cotton wool prevents droplets of solution, that may be thrown out of the solution by the effervescence, from escaping. At the same time, it allows the carbon dioxide to pass through.

1.

Reaction	Reaction equation	Method to monitor rate
1	$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$	Change in pressure or volume
2	$10\text{FeSO}_4 \text{ <pale green> } + 2\text{KMnO}_4 \text{ <purple> } + 8\text{H}_2\text{SO}_4 \rightarrow \text{K}_2\text{SO}_4 + 2\text{MnSO}_4 \text{ <pale pink> } + 5\text{Fe}(\text{SO}_4)_3 \text{ <yellow brown> } + 8\text{H}_2\text{O}$	Colorimetry
3	$\text{C}_2\text{H}_4 + \text{Br}_2 \text{ <red brown> } \rightarrow \text{C}_2\text{H}_4\text{Br}_2 \text{ <colourless> }$	Colorimetry
4	$\text{BrO}_3^- + 5\text{Br}^- + 6\text{H}^+ \rightarrow 3\text{Br}_2(\text{l}) \text{ <brown> } + 3\text{H}_2\text{O}(\text{l})$	Colorimetry/ Conductivity
5	$\text{NH}_4\text{Cl} + \text{NaOH} \rightarrow \text{NH}_3 + \text{NaCl} + \text{H}_2\text{O}$	Change in pressure or volume / Change in mass
6	$5\text{H}_2\text{O}_2(\text{aq}) + 2\text{HIO}_3(\text{aq}) \text{ <yellow> } \rightarrow 5\text{O}_2(\text{g}) + \text{I}_2(\text{aq}) \text{ <blue black> } + 6\text{H}_2\text{O}(\text{l})$ (colour oscillates between yellow and blue black)	Colorimetry/ Conductivity

Experimental techniques for measuring the rate of reaction:

- Colorimetry
- Change in pressure or volume
- Change in mass
- Conductivity
- Clock reaction



1. Reaction 1

Change in pressure or volume can be used as both the reactants and products are gaseous.

Reaction 2

Colorimetry can be used to monitor the changes in colour. As the reaction starts the reaction mixture is predominantly purple and this fades to a very pale brown colour as the reaction proceeds.

Reaction 3

Changes in colour can be monitored using a colorimeter.

Reaction 4

Colorimetry can be used to follow the appearance of the brown colour.

There are 12 moles of ions on the right side of the equation (reactant side) and no ions on the left side, making it easy to monitor the rate of the reaction using conductivity values.

Reaction 5

Both change in pressure or volume and change in mass can be used as ammonia gas is produced.

Reaction 6

This reaction, the Briggs Rauscher reaction, is an oscillating reaction that involves a colour change from pale yellow to blue black and then back to yellow again. These changes can be monitored using colorimetry or can be measured as a clock reaction using a stopwatch to measure the periodic change in colour.

There are no ions on the product side, only oxygen gas, aqueous iodine and water but the reactant HIO_3 exists as H^+ and IO_3^- ions in solution. The decrease in ions from left to right can be monitored by measuring the conductivity of the reaction mixture.

The concept diagram in **Figure 14** links key concepts for measuring reaction rates.

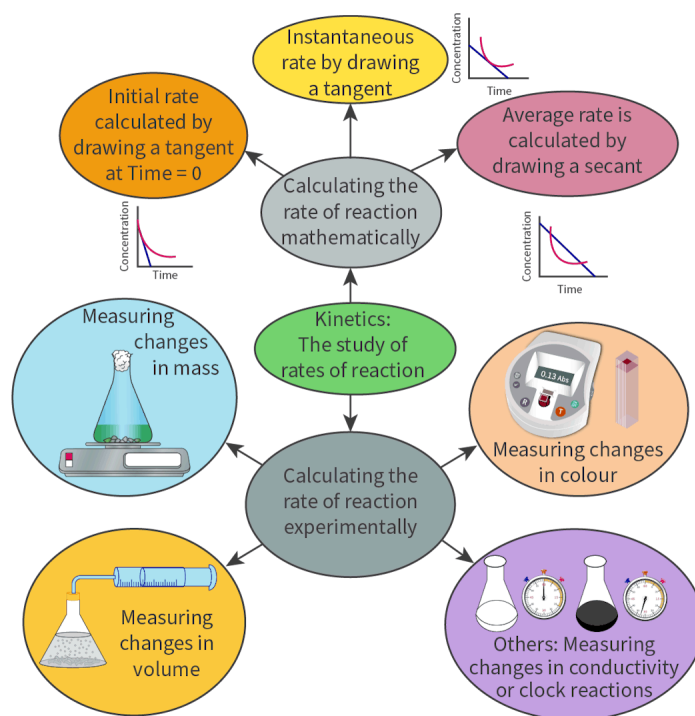


Figure 14. Measuring the rate of reaction.

More information for figure 14

The diagram illustrates the processes and concepts involved in measuring reaction rates. At the center, a gray oval labeled "Calculating the rate of reaction mathematically" connects to multiple surrounding elements.

To the top, a yellow oval states "Instantaneous rate by drawing a tangent," with a small graph showing a curve with tangent. On the top left, an orange oval says "Initial rate calculated by drawing a tangent at Time = 0," alongside a curve graph emphasizing tangency. Top right, a purplish pink oval notes "Average rate is calculated by drawing a secant," with a corresponding curve and secant line on the graph.

Directly below the central oval, a green oval reads "Kinetics: The study of rates of reaction."

On the bottom left, a light blue circle shows an image of a flask placed on a scale with the text "Measuring changes in mass." Further below, a yellow oval labeled "Measuring changes in volume" depicts a syringe setup linked to a flask.

On the bottom right, an orange oval with a spectrophotometer illustration is labeled "Measuring changes in colour." Lastly, on the lower right, a purple circle labeled "Others: Measuring changes in conductivity or clock reactions" depicts flasks and stopwatches.

[Generated by AI]

Factors affecting the rate of reaction

Learning outcomes

By the end of this section you should be able to:

- (R2.2.2) Explain the relationship between the kinetic energy of the particles and the temperature in kelvin, and the role of collision geometry.
- (R2.2.3) Explain the effect of changing conditions on the rate of reaction.

Reaction dynamics: collision theory

For millions of years, natural mechanisms have controlled numerous reactions from digestion and respiration in living organisms to the reactions occurring in the ozone layer surrounding the planet Earth. Humanity too has learned to control chemical reactions to their favour, like slowing down the rotting of food by using preservation techniques and carrying out controlled explosions for building demolition. In order to control reactions, it is very important to know how reactions take place and what factors affect the rate of reaction. Collision theory helps explain the dynamics of chemical reactions. What factors can you think of that may affect everyday reactions around us?

Making connections

You have studied kinetic molecular theory in sections S1.1.2a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/>) and S1.1.3a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/temperature-and-kinetic-energy-id-43068/>). The theory is based on the fact that all particles move as they possess kinetic energy. The temperature in kelvin is a measure of the average kinetic energy of the particles.

Figure 1 shows a commercial cold storage facility where fruits are held prior to shipping. The fruit needs to be kept at optimum temperature and humidity conditions, cold enough to slow spoilage but without allowing the produce to freeze.



Figure 1. Commercial cold storage facility.

Credit: dusanpetkovic, Getty Images

Kinetic energy and temperature

Increasing the temperature means an increase in the average kinetic energy of the particles of a substance. When we heat a substance, we raise the average kinetic energy of that substance. This results in the particles moving faster and colliding more often. **Video 1** shows how the collision frequency of the particles of a gas increases as the temperature increases, given the volume is constant.

Kinetic Energy in a Gas



Video 1. The influence of temperature on particle motion.

Chemical reactions occur when the bonds in the reactants are broken and new bonds are formed in the products. In order to break chemical bonds, the reacting particles must collide with each other, but not all collisions will result in a chemical reaction. For a collision to lead to a chemical reaction, it must have enough energy for the bonds in the reactants to break and the particles must collide with the right orientation. If all the collisions in a gaseous mixture caused a reaction then the reaction would be over in microseconds. But that does not happen. Just like oxygen and nitrogen molecules exist in the air without reacting, reactions only proceed when there is a successful collision.

Concept

The absolute temperature of a gas is directly proportional to the average kinetic energy of the particles. This means a sample of gas with a temperature of 100 K has twice the average kinetic energy of a sample at 50 K.

Energy of particles

As the temperature increases, the average kinetic energy of the particles also increases. As it is the average kinetic energy, this means not all the particles have the same amount of energy. So that when particles collide with each other not all collisions are successful, even if the orientation is correct. At increased temperatures, a greater fraction of molecules will have the energy necessary for overcoming the repulsions between reactants and breaking the bonds in the reactants so that the reaction can occur.

Collision geometry

Collisions in a reaction mixture occur randomly. This means that the particles collide in many different orientations. Only the collisions that have the right orientation are successful collisions and lead to a reaction, given that the amount of energy is sufficient. That is the particles must be facing each other the right way when they collide. **Video 2** explains this.

Collision Theory



Video 2. The geometry of collisions.

Study skills

The explanation of what is happening in particulate models regarding the collision theory needs to be very precise. It is the frequency of collisions that increases, that is, the number of collisions per second and not just the number of collisions.

Aspect: Theories

Collision theory helps explain the effects of several factors on the rate of reaction, although it is not based on actual observations of collisions between particles. It is actually based on the kinetic molecular theory. This is what theories do: they include assumptions from other theories and help scientists make useful predictions.

Factors affecting the rate of a reaction

From the kinetic molecular theory, we know anything that will increase the frequency of successful collisions will increase the rate of reaction. The following factors affect the rate of a reaction.

Temperature

Increasing the temperature increases the average kinetic energy of the particles. This results in an increase in collision frequency and a greater portion of particles now have sufficient energy needed for a reaction. We know for a fact that the frequency of collision is proportional to the square root of the increase in temperature. It turns out that the effect of an increase in collisions on the reaction rate is minor. It is the increase in the fraction of molecules with high energy, that has a bigger effect on reaction rate.

$$\text{rate of reaction} = \text{collision frequency} \times \text{fraction of effective collisions}$$

For example, for any given reaction:

$$\text{rate} = (1000 \text{ collisions s}^{-1}) \times (1 \text{ successful collision every } 1000 \text{ collisions})$$

$$\text{rate} = 1 \text{ successful reaction s}^{-1}$$

As increasing temperature increases both collision frequency and the fraction of successful collisions (due to an increase in energy) the rate also increases.

Many reactions double their reaction rate for every 10 °C increase in temperature. The effect of temperature on the reaction rate can be seen in **Figure 2**. Which curve do you think is at the higher temperature, X or Y? Note that the average rate of reaction over the course of the reaction will be the same for both temperatures - can you explain why?

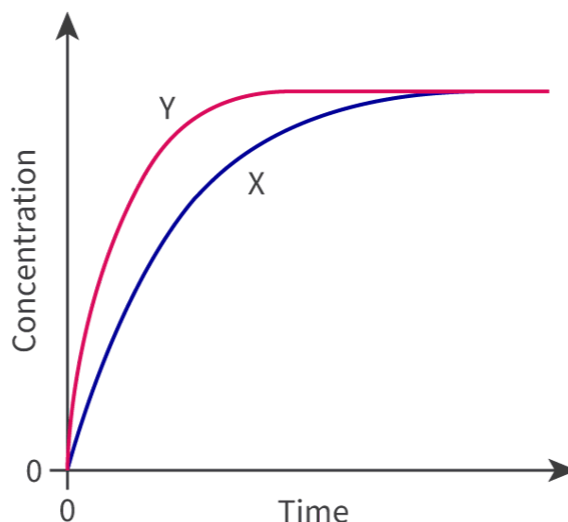


Figure 2. The effect of temperature on the rate of reaction.

 More information for figure 2

The image is a graph depicting the effect of temperature on the reaction rate, with concentration on the Y-axis and time on the X-axis. There are two curves, labeled X and Y. Curve Y reaches near peak concentration faster than curve X, indicating a higher reaction rate. Both curves ultimately plateau at the same concentration level, illustrating that although Y reaches its peak concentration quicker, the final concentration is the same for both reactions. This implies that the reaction at higher temperature completes faster but achieves the same equilibrium concentration as the reaction at lower temperature.

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Concentration

As we increase the concentration the frequency of collision increases, and this results in more successful collisions between the particles of reactants and products. An increase in frequency of collisions means the rate increases. We know that:

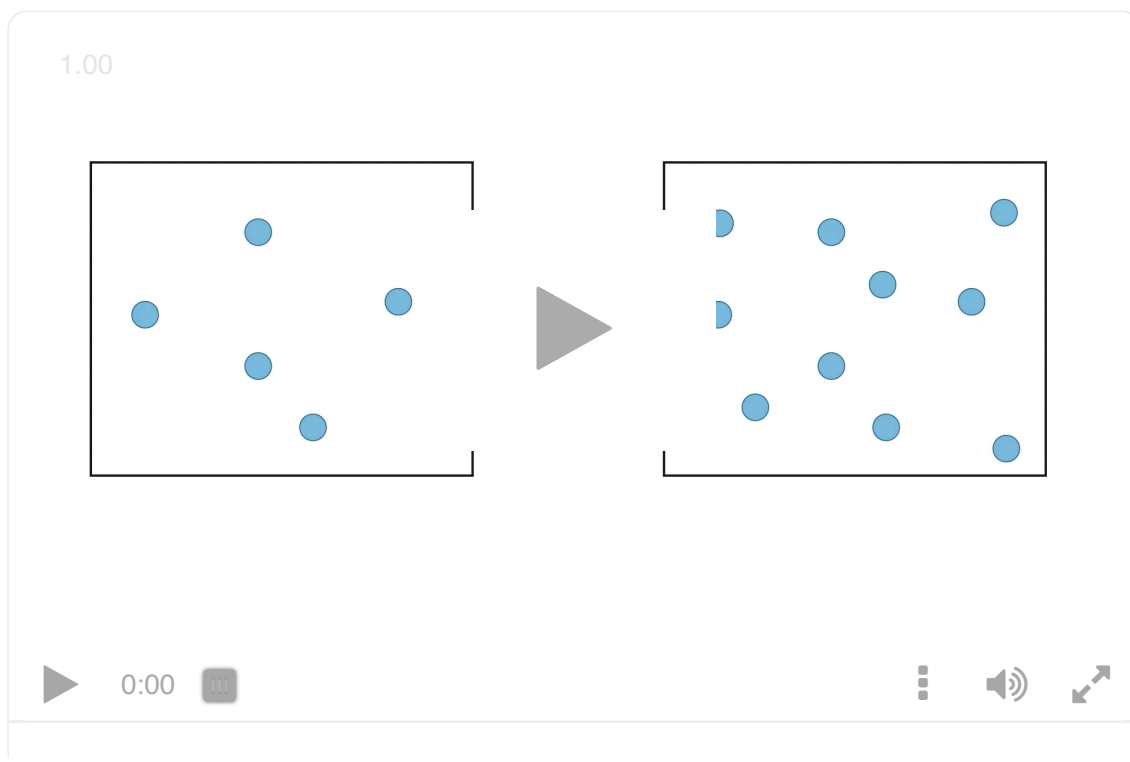
$$\text{rate of reaction} = \text{collision frequency} \times \text{fraction of effective collisions}$$

Although collision frequency is increasing, the fraction of effective collisions (with sufficient energy) is not, as there is no increase in energy. This means the effect of an increase in concentration does not increase the rate as much as an increase in temperature would because there is no increase in the fraction of particles with sufficient energy.

Concept

Remember an increase in concentration only affects a reaction where the reactants are in aqueous solution. There is no effect on the collision frequency of solid reactants.

The following simulation shows how collision frequency increases as concentration increases.



Interactive 1. The Effect of Concentration on the Collision Frequency Between Molecules.

 More information for interactive 1

An interactive video illustrates two containers containing spherical particles. The container on the left has a lower concentration, with about five particles, while the container on the right has a higher concentration, with about 10 particles. The particles move randomly within each container.

Collisions between particles occur less frequently in the left container due to the lower number of particles, whereas in the right container, more frequent collisions occur due to the higher concentration.

This video helps users understand how an increase in concentration leads to a higher collision frequency, while a decrease in concentration has less collision frequency.

The two graphs in **Figure 3** illustrate how the rate of reaction is affected by increasing the concentration of the hydrochloric acid when it reacts with calcium carbonate. Note that when the hydrochloric acid is the limiting reagent both the yield and rate increase (Graph A). When calcium carbonate is the limiting reagent, there is no effect on the amount of carbon dioxide produced and only the rate increases (Graph B).

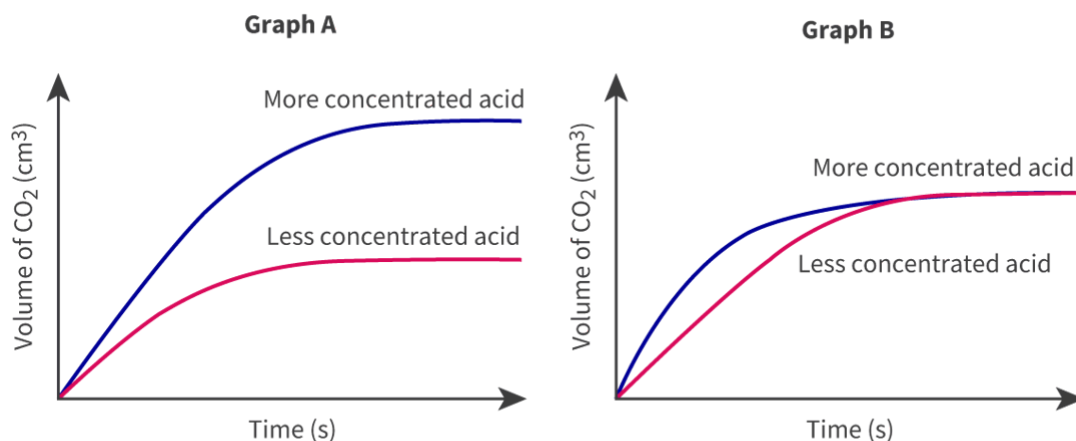


Figure 3. The effect of concentration of reactant on the rate of reaction.

 More information for figure 3

The image contains two graphs labeled as Graph A and Graph B.

Graph A: The X-axis is labeled "Time (s)", and the Y-axis is labeled "Volume of CO₂ (cm³)". The graph displays two curves representing different concentrations of hydrochloric acid. The curve for "More concentrated acid" rises sharply and levels off at a higher volume of CO₂ than the curve for "Less concentrated acid". This indicates that when the hydrochloric acid is the limiting reagent, both the yield and reaction rate increase.

Graph B: Similarly, the axes are the same as in Graph A, but the behavior of the curves is different. Both curves eventually level off at the same volume of CO₂ but reach this point more quickly for "More concentrated acid" compared to "Less concentrated acid". This shows that when calcium carbonate is the limiting reagent, the reaction rate increases with acid concentration, but the total amount of CO₂ remains unchanged.

[Generated by AI]

Concept

Remember the rate of reaction changes as the reaction proceeds. This is because the concentration of the reactants is changing.

Video 3 shows how the effect of concentration can be investigated for the hydrochloric acid and calcium carbonate reaction.

Effect of Concentration on the Rate of Reaction | Chemistry ...



Video 3. Investigating the effect of concentration on the hydrochloric acid and calcium carbonate reaction.

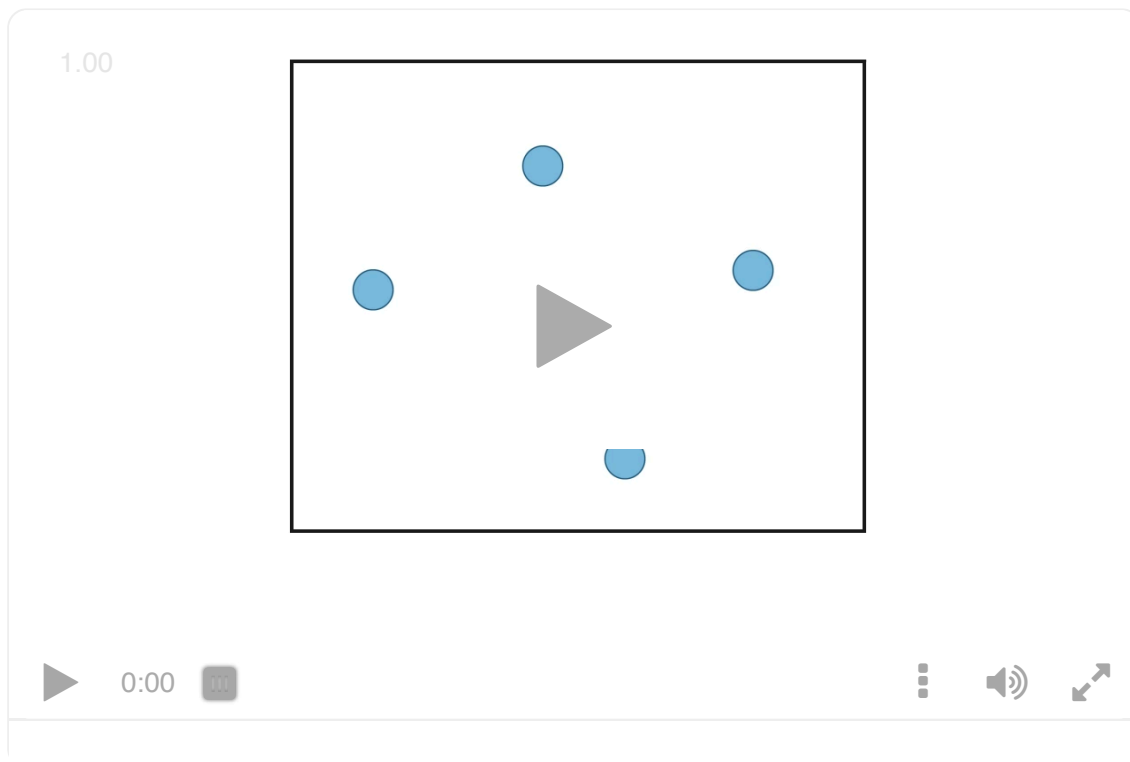
Practical skills

- **Inquiry 1:** Exploring and designing
- **Skill:** Identify and justify the choice of dependent, independent and control variables

Remember the rate of reaction depends on several factors. When investigating one factor the other factors must be controlled. For an experiment investigating the effect of concentration on the rate of reaction, the temperature and surface area of the reactants must remain constant. Sometimes such a reaction is carried out in an electric water bath to maintain temperature.

Pressure

For gaseous reactions increasing the pressure increases the rate of reaction. This is because an increase in pressure, whether it is due to compressing the gas or increasing the number of molecules, effectively increases its concentration. This leads to an increase in the frequency of collisions, as seen in **Video 4**.



Video 4. The Effect of Pressure on Reaction Rate.

 [More information for video 4](#)

An interactive video illustrates a container filled with five gas particles moving in a random motion. As the video progresses, the pressure inside the container increases, making the particles compress into the reducing volume. This causes them to move more randomly and collide more frequently. When the pressure inside the container reduces, the frequency of collisions also gets reduced.

This video helps users understand how increasing pressure leads to a higher collision frequency between gas particles.

Surface area

In heterogeneous reactions, where the reactants are in different phases, the reaction can only occur when the two phases meet and collisions between reactant particles become possible. For example, in a reaction between hydrochloric acid and marble chips, the reaction proceeds faster when the marble chips are powdered as opposed to using chips. This is because decreasing the size

of the particles increases their surface area and more of the calcium carbonate particles make contact with the solution, therefore increasing the frequency of collisions, as seen in **Figure 4**.

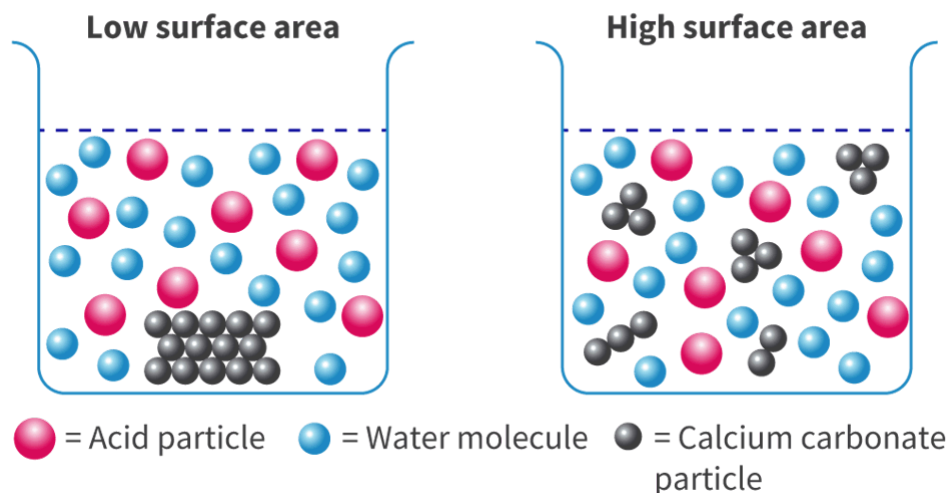


Figure 4. The effect of surface area on the rate of reaction.

 More information for figure 4

The image depicts two beakers side by side, illustrating the effect of surface area on the reaction rate. Both beakers contain a mixture of acid particles, water molecules, and calcium carbonate particles.

On the left, the beaker is labeled "Low surface area" and shows larger clusters of calcium carbonate particles surrounded by acid particles and water molecules. This results in fewer contact points between reactants.

On the right, the beaker is labeled "High surface area" and features smaller, scattered calcium carbonate particles intermixed more uniformly with acid particles and water molecules. This configuration increases contact points and potential collisions among reactants.

The key at the bottom of the image identifies the particles: pink circles represent acid particles, blue circles denote water molecules, and grey circles signify calcium carbonate particles.

[Generated by AI]

Increasing the surface area of contact between reactants increases the rate of reaction. The daily life consequences of increasing surface area have also been discussed in The big picture S2.2.0 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/>).

Video 5 shows a laboratory demonstration of the effect of surface area on the reaction rate.

The effect of surface area on reaction rate

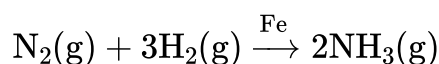


Video 5. Surface area and reaction rate in the laboratory.

Catalyst

A catalyst is a substance that changes the rate of a chemical reaction but itself remains unchanged during the reaction. Most catalysts work by providing an alternative pathway to the reaction that requires a lower energy. This means that without increasing the temperature or concentration we can speed up the chemical reaction as a lower energy alternative route becomes available for the reaction to take place.

Catalysts are very specific for particular reactions. Different catalysts catalyse different reactions. The production of ammonia by the Haber process uses an iron (Fe) catalyst. As the catalyst does not take part in the reaction, it is not included in the actual reaction but is written on top of the arrow.



Biological reactions are controlled by catalysts called enzymes. Thousands of enzymes exist as they are each specific for a particular reaction. For example, the digestive enzyme lactase breaks down the lactose sugar in milk (**Figure 5**). Some people's bodies do not produce enough lactase and they are said to be lactose intolerant. Without lactase, the lactose cannot be broken down during digestion and this results in unpleasant digestive symptoms like pain and discomfort.

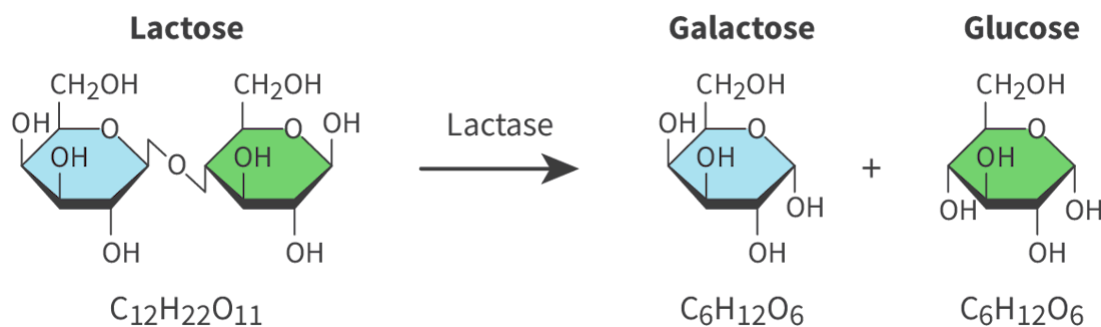


Figure 5. The enzyme catalysed breakdown of lactose into glucose and galactose.

More information for figure 5

The image is a chemical diagram illustrating the enzymatic breakdown of lactose into glucose and galactose. On the left, the lactose molecule is depicted with its chemical formula $C_{12}H_{22}O_{11}$, showing its structure with two ring-like formations. To the right, there's an arrow labeled "Lactase" indicating the enzymatic process. On the right, two separate molecules are shown: galactose and glucose, each with the chemical formula $C_6H_{12}O_6$. The diagram visually represents the biochemical reaction facilitated by the enzyme lactase, highlighting the conversion of lactose into two simpler sugar molecules, glucose and galactose.

[Generated by AI]

Catalysts can also play a significant role in improving society, such as catalytic converters (**Figure 6**) that use precise metals such as platinum and palladium to remove harmful gases from vehicle exhausts.

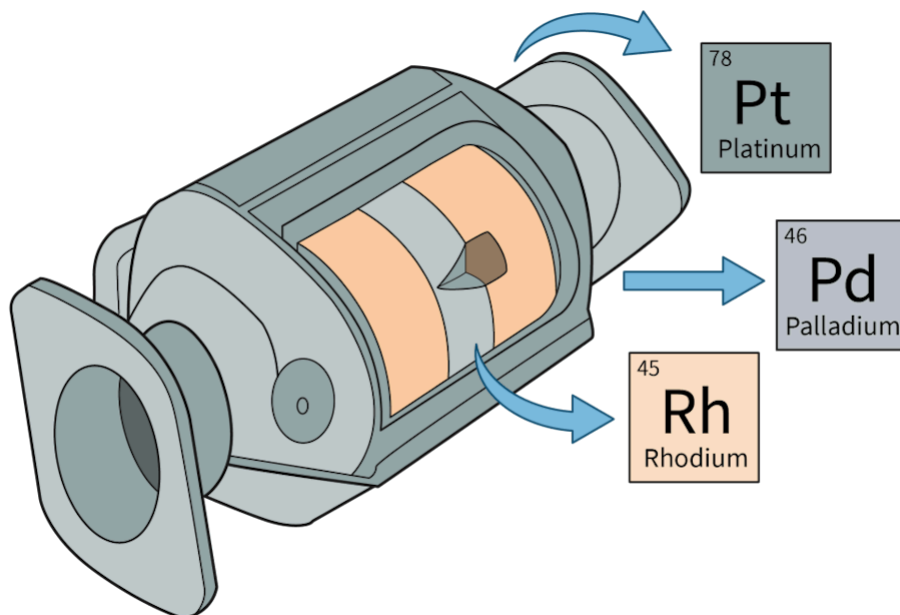


Figure 6. A catalytic convertor.

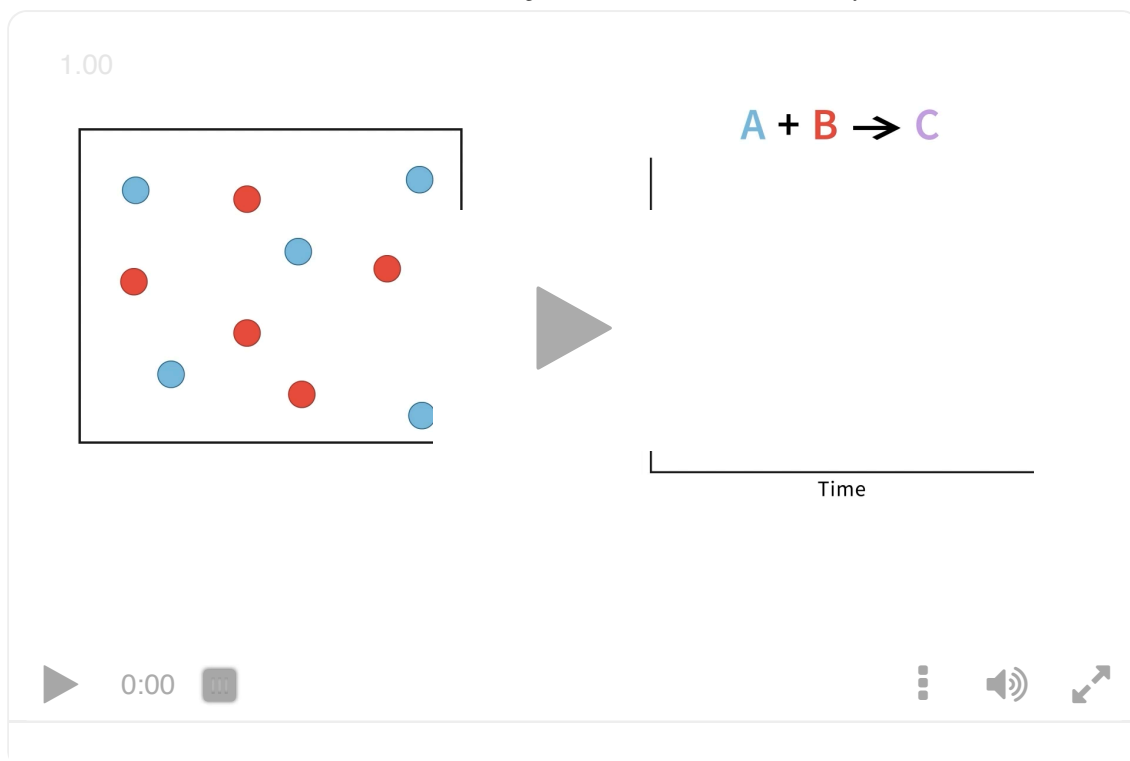
 More information for figure 6

The image shows an illustrated diagram of a catalytic converter. The converter is cylindrical with a slightly narrowing section toward the end. It contains arrows pointing at it that are labeled with elements and their details: an arrow labeled 'Pt' for platinum with an atomic number 78, another labeled 'Pd' for palladium with atomic number 46, and one labeled 'Rh' for rhodium with atomic number 45. These arrows indicate that these metals are components inside the catalytic converter, which is instrumental in converting harmful gases into less harmful emissions.

[Generated by AI]

Investigating rates of reaction

1. When you run **Interactive 2** the reaction in the left-hand box proceeds as the two reactants combine to create the product.
2. As the reaction proceeds, witness what happens on the graph on the right-hand side.



Interactive 2. The Plotting of Product Concentration vs. Time Graph to Monitor a Reaction.

More information for interactive 2

An interaction video demonstrates how the amount of product changes over time during a reaction.

On the left-hand side, a box contains five A particles and five B particles, which act as reactants. These reactants collide to form a new product consisting of five C particles. On the right-hand side, a graph plotting the amount of product forms versus time depicts data for the following reaction:



As the reactants start to form the products, a line in the graph starts at the origin and increases linearly over time.

This video helps users understand how product formation progresses over time.

Making connections

You can read more about how catalysts work in [sections R2.2.4—5](#)

([https://app.kognity.com/study/app/preview-p/sid-426-cid-](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/maxwell-boltzmann-distribution-curves-id-44782/)

[753301/book/maxwell-boltzmann-distribution-curves-id-44782/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/maxwell-boltzmann-distribution-curves-id-44782/)) and [S3.1.8](#)

([https://app.kognity.com/study/app/preview-p/sid-426-cid-](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/)

[753301/book/content-to-come-2-id-43595/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/)).

Theory of Knowledge

There is evidence dating back to 10 000 BCE that ancient societies preserved milk, vegetables and fruits using fermentation. Fermentation is a process where the enzymes in microbes, such as yeast and bacteria, break down sugars and convert them to alcohol and acids to prevent spoilage when refrigeration was not yet possible. Foods such as yogurt, pickles, bread and tea have been made using natural fermentation methods for thousands of years. Before globalisation, different societies around the world found ways to ferment similar food items independently of each other, well before the knowledge of enzymes. What role does intuition play in the development of scientific processes? Does the fact that the knowledge of enzymes was not obtained by scientific experiment but by the intuition of societies for survival make the knowledge less valid?

Complete the section questions below and then click 'Next' to see the activity for this section.

Maxwell-Boltzmann distribution curves

Learning outcomes

By the end of this section you should be able to:

- (R2.2.4) Sketch Maxwell-Boltzmann energy distribution curves to explain the effect of temperature on the probability of successful collisions.
- (R2.2.5) Sketch and explain energy profiles with and without catalyst for endothermic and exothermic reactions.
- (R2.2.5) Construct Maxwell-Boltzmann distribution curves to explain the effect of different values of E_a on the probability of successful collisions.

Activation energy, E_a

When you turn on the methane gas in your stove, it does not start burning as soon as you turn the knob. You need to ignite the gas for the reaction to start, once that is done the reaction proceeds spontaneously. The precise amount of energy needed to ignite the gas is called the activation energy (E_a) for the combustion of methane.

You have seen energy profile diagrams in [section R1.1.3](https://app.kognity.com/:sectionlink:138726)  (<https://app.kognity.com/:sectionlink:138726>). For an endothermic reaction the products are at a higher energy than the reactants as shown in **Figure 1**.

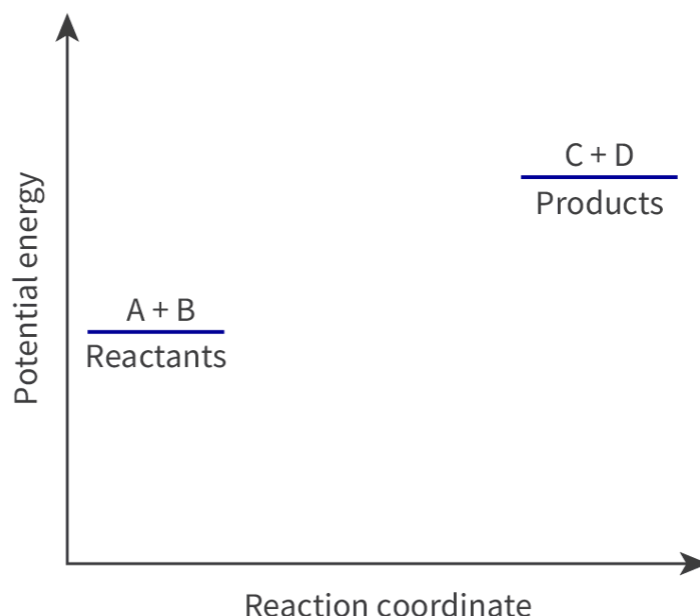


Figure 1. An energy profile diagram for an endothermic reaction.

 More information for figure 1

This is an energy profile diagram for an endothermic reaction. The X-axis represents the reaction coordinate, while the Y-axis represents potential energy. The diagram shows two horizontal lines. The first line, labeled "A + B Reactants," is positioned lower on the Y-axis, indicating a lower potential energy level. The second line, labeled "C + D Products," is positioned higher on the Y-axis, indicating a higher potential energy level than the reactants. This illustrates the typical energy increase seen in an endothermic reaction, where the products have higher energy than the reactants.

[Generated by AI]

If we draw an energy profile diagram to show what actually happens to the energy during the reaction it would appear like this.

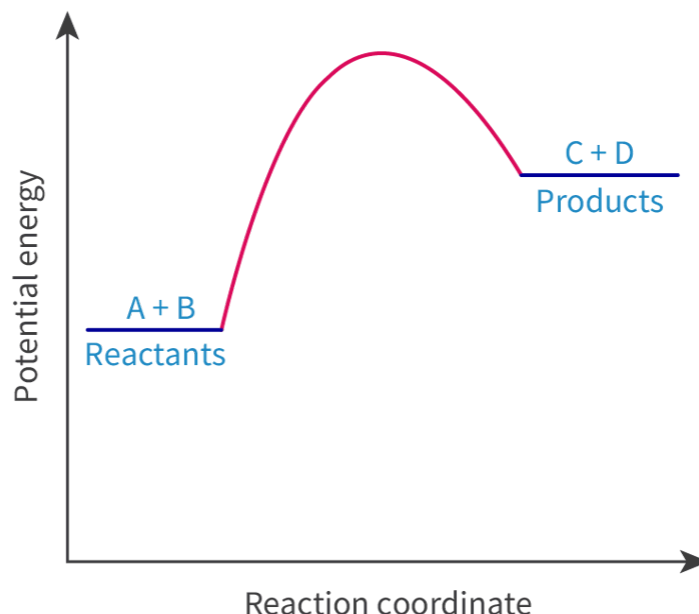


Figure 2. An energy profile diagram for an endothermic reaction showing the bump for activation energy.

 More information for figure 2

The image is a diagram representing an energy profile for an endothermic reaction. It is a graph plotting potential energy on the vertical axis against the reaction coordinate on the horizontal axis. The diagram illustrates a curve starting from a lower energy level indicating the reactants labeled as 'A + B'. Then, it rises to form a peak, which represents the activation energy — the threshold energy required for the reaction to occur. After the peak, the curve descends to another energy level representing the products labeled as 'C + D'. The peak is higher than both the starting and ending points, showing that the reaction requires input energy to reach the transition state before forming products, which signifies an endothermic process. The axes are marked with 'Potential Energy' on the vertical and 'Reaction Coordinate' on the horizontal.

[Generated by AI]

What do you think the hump in the energy represents? The activation energy of a reaction is the minimum amount of energy required by reactant particles to form product particles. We know that for a collision to be successful it must have sufficient kinetic energy. This energy goes into breaking the bonds in the reactants so that the reaction may proceed. When this energy is available the reactants achieve a transition state from which they can then form products. So only the particles that have energy greater than or equal to the activation energy will undergo successful collisions. Other particles will still collide, but the collisions will not lead to a reaction.

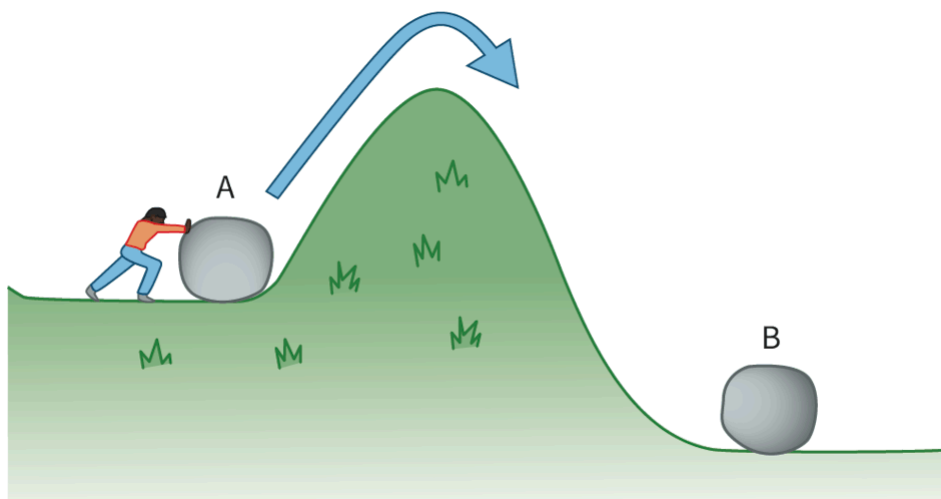


Figure 3. The activation energy barrier.

 More information for figure 3

The image is a diagram depicting a person labeled 'A' pushing a rock up a grassy hill with a hump. The hill has a green surface with patches of grass. The person is positioned at the lower left side, exerting force on the rock to move it upward over the hump. An arrow is drawn over the hill, indicating the direction of movement from left to right, going over the hump and down towards another rock labeled 'B' at the bottom right of the hill. This illustrates the concept of an activation energy barrier in a chemical reaction where an initial push is needed to overcome a resistance before achieving a transition state.

[Generated by AI]

Concept

A transition state is a species that forms when a reaction has the activation energy needed to break the bonds in the reactants. The transition state is formed midway between the reactants and products, with partially broken and partially formed bonds. This makes the transition state unstable, and it quickly rearranges itself to form the products.

The energy profile diagrams in **Figure 4** show the activation energy for an endothermic and exothermic reaction. The transition state is also shown.

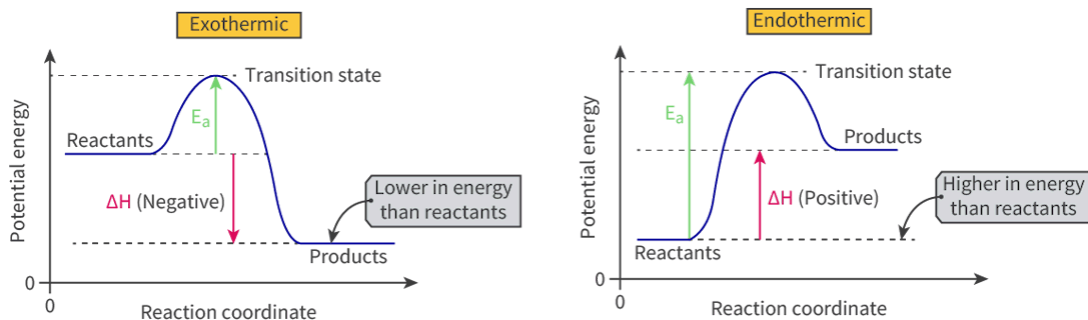


Figure 4. Energy profile diagrams for exothermic and endothermic reactions.

 More information for figure 4

The image contains two energy profile diagrams illustrating exothermic and endothermic reactions. Each graph plots potential energy against reaction coordinate.

1. Exothermic Reaction Diagram (Left):

2. The X-axis represents the reaction coordinate, and the Y-axis represents potential energy.
3. The curve starts at the reactants' energy level, rises to a peak identified as the transition state, and then falls to a lower energy level for the products.
4. The activation energy (E_a) is shown as a green upward arrow from reactants to the peak of the transition state.
5. The change in enthalpy (ΔH) is negative, indicated by a red downward arrow from reactants to products.
6. A label notes that products are lower in energy than reactants.

7. Endothermic Reaction Diagram (Right):

8. Similarly, the X-axis is the reaction coordinate, and the Y-axis shows potential energy.
9. The curve begins at the reactants' energy level, rises to a transition state, and then decreases to a higher energy level for the products.
10. The activation energy (E_a) is depicted as a green upward arrow from the reactants to the peak.
11. The change in enthalpy (ΔH) is positive, marked by a red upward arrow from reactants to products.

12. A label indicates that products are higher in energy than reactants.

[Generated by AI]

Energy profile diagrams for a catalysed and uncatalysed reactions

Figure 5 shows the profile diagram for an exothermic and endothermic reaction with and without a catalyst.

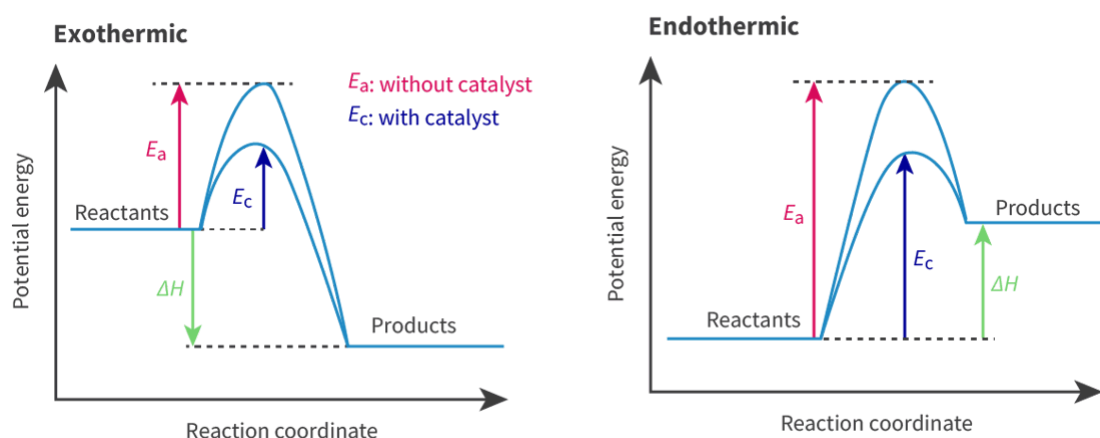


Figure 5. An exothermic and endothermic reaction profile diagram showing a catalysed and uncatalysed reaction.

 More information for figure 5

The diagram illustrates two reaction profiles side by side: one for an exothermic reaction and one for an endothermic reaction. Each profile compares catalyzed and uncatalyzed reactions.

On the left is the exothermic reaction graph. The X-axis is labeled 'Reaction coordinate,' and the Y-axis is labeled 'Potential energy.' The graph starts with 'Reactants' at a relatively high energy level, peaking at the transition state. The peak labeled as ' E_a ' indicates the activation energy without a catalyst. A lower peak labeled ' E_c ' shows the reduced activation energy with a catalyst. The graph descends as it moves to 'Products,' showing a decrease in potential energy, marked by ' ΔH ' indicating an overall energy release.

On the right is the endothermic reaction graph. It has similar axis labels to the exothermic graph. The 'Reactants' start at a lower potential energy. There are two peaks, with ' E_a ' showing the higher activation energy without a catalyst, and ' E_c '

displaying the lower activation energy with a catalyst. The potential energy increases in the products, highlighted by ' ΔH ' indicating energy absorption.

Overall, both diagrams illustrate how catalysts lower the activation energy needed for reactions, affecting energy absorption and release differently in exothermic and endothermic processes.

[Generated by AI]

The catalyst provides an alternative route with lower activation energy; the profile diagrams in **Figure 5** show this. The activation energy for the catalysed reaction is much lower than the activation energy of the uncatalysed reaction.

Study skills

- Note that the activation energy for the endothermic reaction spans the entire height of the peak and not just the hump.
- Remember that ΔH remains the same regardless of whether a catalyst is used or not. The $E_{\text{cat}} < E_a$ hence the rate of reaction is faster in both the catalysed exothermic and endothermic reactions

Maxwell-Boltzmann distribution curves

Temperature is a measure of the average kinetic energy of the molecules. In a sample at a given temperature, not all particles will possess the same amount of energy. This means that at any given temperature some fraction of the molecules will possess very high energy and will move very fast. Another fraction will have very low energy and will move very slowly. But most molecules have energy between these two extremes. The Maxwell-Boltzmann distribution curve shows the distribution of these energies (**Figure 6**). The y-axis on the Maxwell-Boltzmann graph can be thought of as giving the number of molecules possessing the given energy. This is very similar to say a distribution of heights graph for the pupils in your classroom. If you plot the range of heights on the (x-axis) and the number of students on the y-axis, the y-axis will give you the number of students falling in a specific height range. In the same way the x-axis on the Boltzmann curve gives the amount of energy that the particles possess.

The shape of the Maxwell-Boltzmann distribution curve is not a bell curve, showing that the distribution of energies is not a normal distribution. In fact, it is shifted to the left, showing that there are more particles with lower energy and fewer with higher energy.

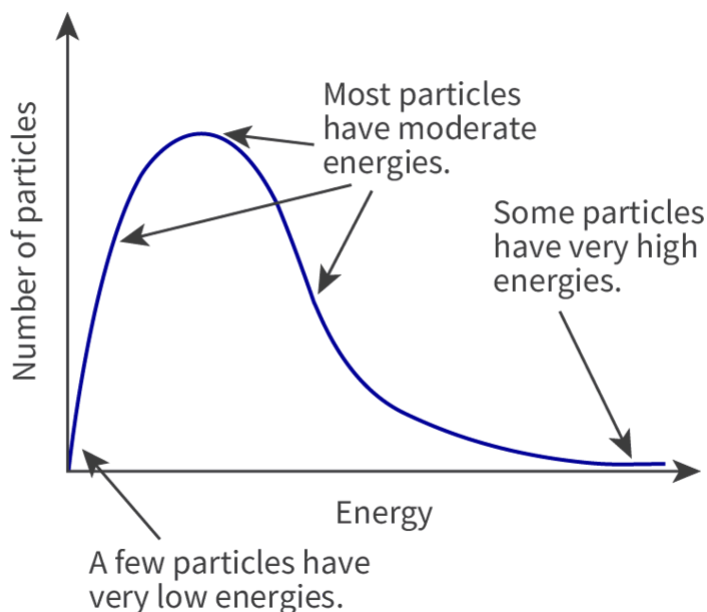


Figure 6. The Maxwell-Boltzmann distribution curve — energy distribution in the molecules of a gas.

Note the area under the curve is a measure of the total number of particles present.

 More information for figure 6

This image is a graph depicting the Maxwell-Boltzmann distribution curve for energy distribution among gas molecules. The X-axis represents 'Energy' and the Y-axis represents 'Number of particles'. The curve illustrates that most particles have moderate energies, peaking in the middle of the graph. There are fewer particles with very high energies towards the right, and fewer with very low energies towards the left. The graph highlights key areas: the left shows "A few particles have very low energies", the peak shows "Most particles have moderate energies", and the right shows "Some particles have very high energies". The curve is not symmetric, showing a skewed distribution to the left, indicating a higher number of particles with lower energy.

[Generated by AI]

The Maxwell-Boltzmann distribution curve and activation energy

When particles collide, they must have energy equal to or greater than the activation energy in order for the reaction to occur. The activation energy can be marked on the Maxwell-Boltzmann curve as shown in **Figure 7**.

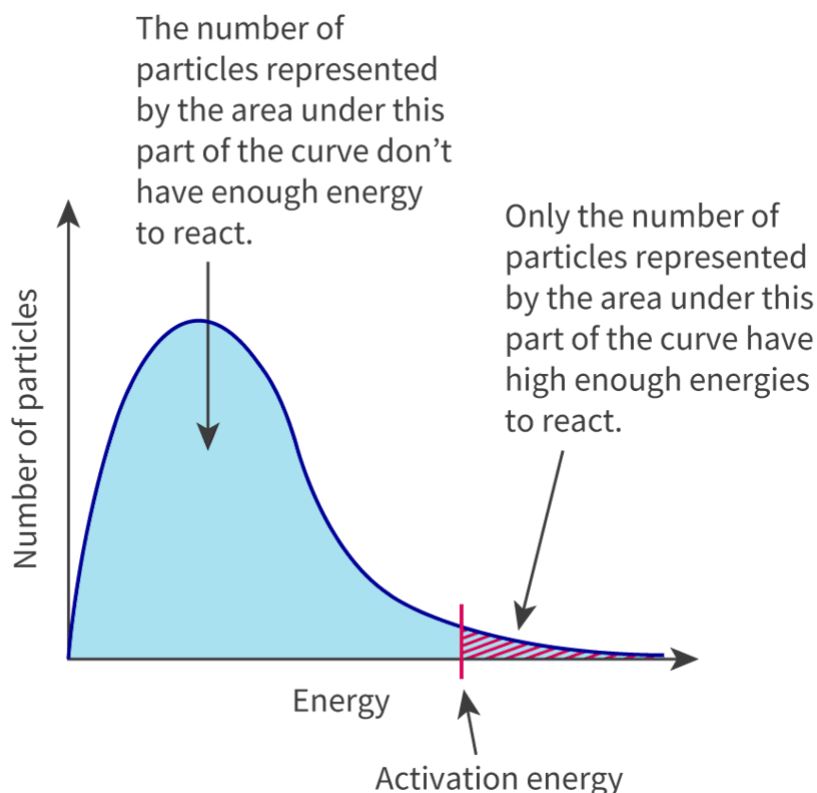


Figure 7. The Maxwell-Boltzmann distribution curve and activation energy.

 More information for figure 7

The image is a graph illustrating the Maxwell-Boltzmann distribution curve related to the kinetic energy of particles. The X-axis is labeled as "Energy," representing the kinetic energy of particles, while the Y-axis is labeled as "Number of particles." The graph depicts a bell-shaped curve that peaks and then declines.

To the left of the curve's peak, an annotation indicates that the number of particles under this part of the curve does not have enough energy to react. To the right of the curve, another annotation points out that only the number of particles under this part of the curve has high enough energies to react.

A vertical line is drawn from the X-axis, intersecting the curve, marking the "Activation energy," which is the minimum energy required for the particles to react upon collision. The area under the curve to the right of this line represents particles with sufficient energy for a reaction, while the area to the left represents particles lacking this energy.

[Generated by AI]

In **Figure 7**, the area under the curve to the right of the activation energy line is the number of particles that have the necessary energy to react on collision. Particles on the left side of the activation energy line do not possess the minimum energy required for a successful collision.

Study skills

Remember!

- The x-axis on the curve is labelled 'Energy' and the y-axis is labelled as 'Number of particles'
- The curve must always start at zero.
- The curve is not symmetrical and tapers off to the right to form a 'tail'.

The Maxwell-Boltzmann curve and the effect of temperature

An increase in temperature increases the reaction rate. This is because there is an increase in the average kinetic energy of the reactant particles and the collision frequency increases. This means that a greater fraction of the particles possess energy equal to or greater than the activation energy. **Figure 8** shows the distribution of kinetic energy at two different temperatures.

Note that the curve for higher temperature has a lower peak height, this is because it has been slightly stretched to the right. You can now see that more particles now possess higher energy for this curve, that is more particles now have energy greater than the activation energy. Although the shape of the two curves is different, the area under the curves remains the same, that is the number of particles is unchanged.

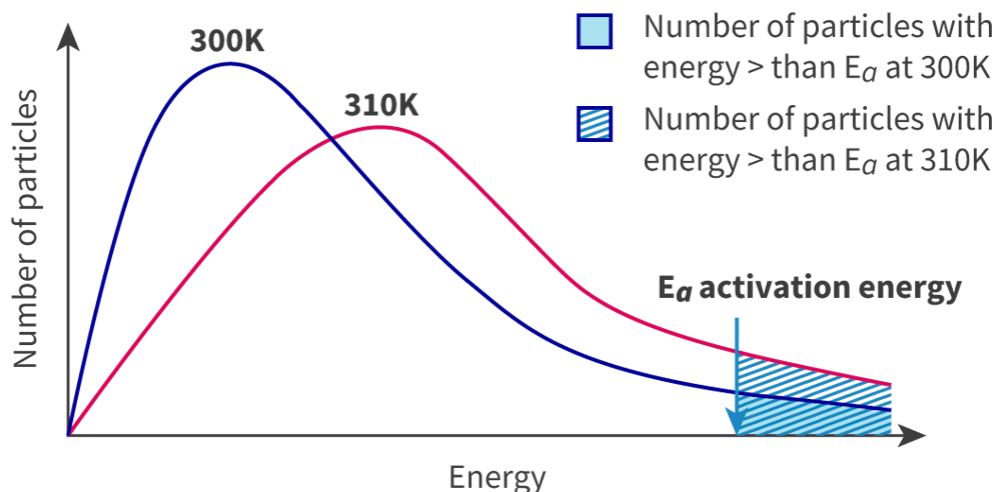


Figure 8. The Maxwell-Boltzmann distribution curve at two temperatures.

 More information for figure 8

The image depicts two Maxwell-Boltzmann distribution curves on a graph. The X-axis represents energy, while the Y-axis represents the number of particles. The two curves illustrate the distribution of particle energy at different temperatures: one for 300K and another for 310K. The 300K curve is colored blue, while the 310K curve is pink. Both curves start from the origin, rise to a peak, and then fall asymptotically back towards the X-axis. The 310K curve has a lower peak height, indicative of increased energy distribution, and is shifted to the right compared to the 300K curve. There are shaded areas under each curve to the right of an indicated activation energy (E_a) line, with the area under the 310K curve having more particles with energy greater than E_a , represented by striped shading.

[Generated by AI]

The shaded area under each curve to the right of the activation energy is the fraction of particles that have sufficient energy to react on collision. The striped area under the higher temperature graph is larger than the blue area under the lower temperature curve. Therefore, at higher temperatures, a greater proportion of the collisions will be successful.

Maxwell-Boltzmann distribution curves and the effect of catalysts

The way that a catalyst speeds up a reaction can be explained using a Maxwell-Boltzmann curve. The presence of the catalyst does not alter the average kinetic energy of the molecules but provides an alternate route with lower activation energy.

This shifts the activation energy line to the left, so that there are now more particles meeting the E_a requirement (**Figure 9**).

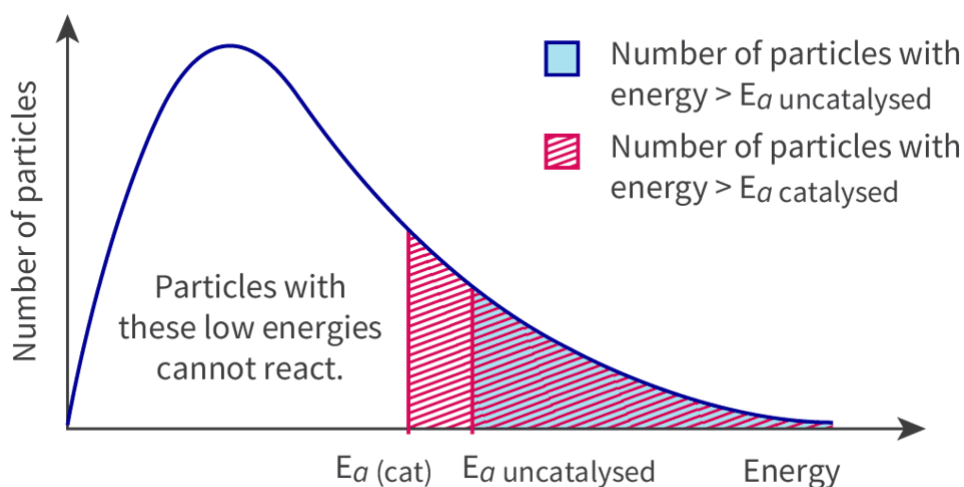


Figure 9. The Maxwell-Boltzmann diagram for a reaction with and without a catalyst.

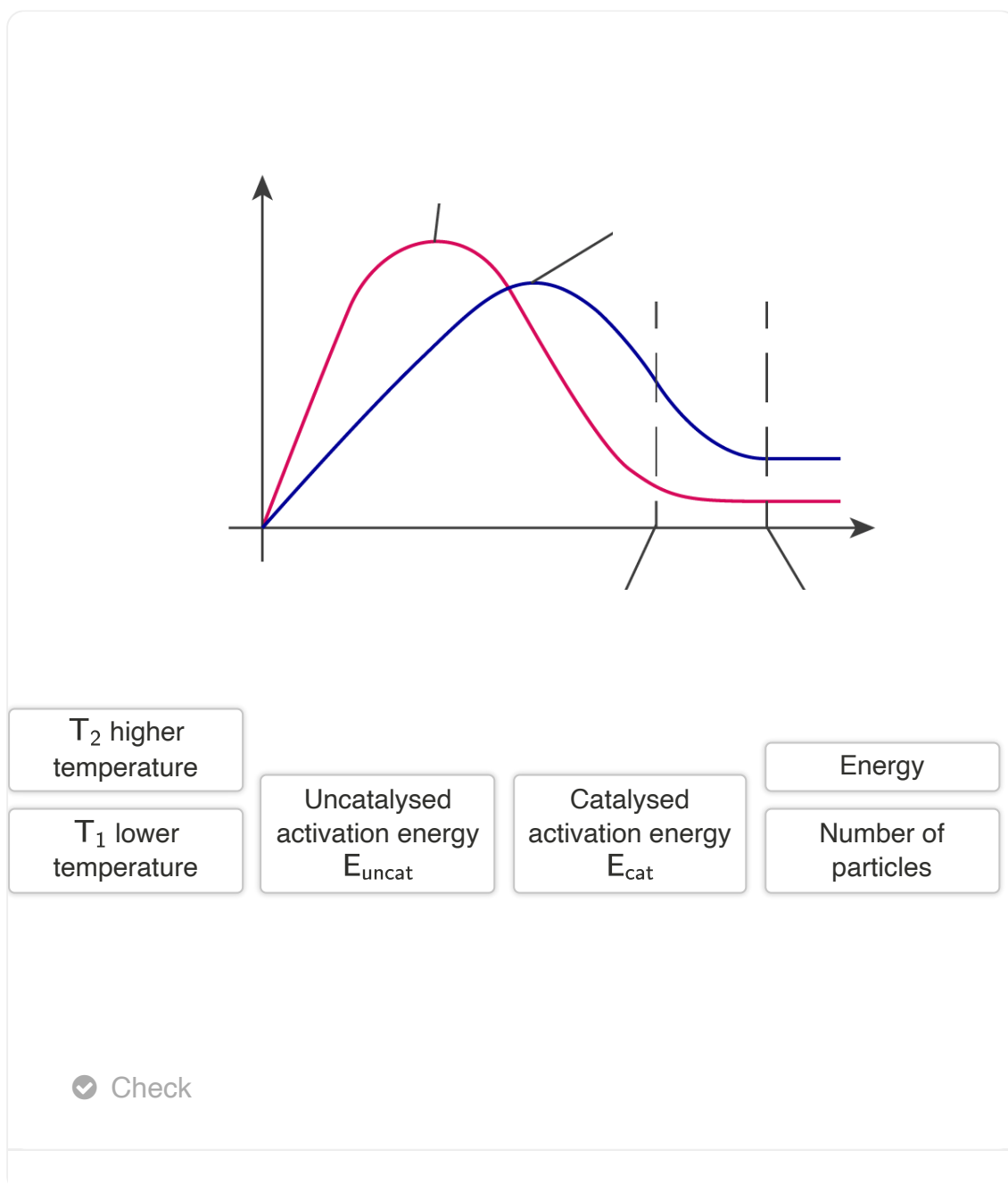
 More information for figure 9

The diagram is a Maxwell-Boltzmann distribution graph. The X-axis represents energy, while the Y-axis represents the number of particles. There are two activation energy lines marked: $E_a(\text{cat})$ and $E_a(\text{uncatalysed})$. To the right of these lines are shaded areas under the curve, illustrating the number of particles with sufficient energy to react. The area to the right of the $E_a(\text{cat})$ is larger, indicating more particles meet the catalysed energy requirement compared to the uncatalysed one. A section in blue shows the number of particles with energy greater than $E_a(\text{uncatalysed})$, and a section in red with stripes shows particles with energy greater than $E_a(\text{catalysed})$. A note on the left indicates particles with low energies cannot react.

[Generated by AI]

The shaded area on the right of each activation energy line is the number of particles that have sufficient energy to react upon collision. Activation energy of the catalysed reaction, $E_{a(\text{cat})}$, is lower than the uncatalysed activation energy, $E_{a(\text{uncat})}$, so the striped area under the curve to the right of the $E_{a(\text{cat})}$ is greater than the area under the curve to the right of the $E_{a(\text{uncat})}$. This shows that a greater proportion of the collisions involving the catalyst will be successful. Therefore, the rate of reaction will be greater in the presence of the catalyst.

Use **Interactive 1** to drag and drop the labels to the correct places.



Interactive 1. The Maxwell-Boltzmann Distribution Curve — Energy Distribution in the Molecules of a Gas.

More information for interactive 1

This interactive Maxwell—Boltzmann distribution graph illustrates how temperature and the presence of a catalyst influence the energy distribution of reacting particles in a system.

- The graph shows two bell-shaped curves, both skewed to the right.
 - Red curve represents particles at a lower temperature (T_1).
 - The blue curve represents particles at a higher temperature (T_2).
- The Y-axis indicates the number of particles.
- The X-axis shows the energy of the particles.
- Two dashed vertical lines near the tail represent:
 - The activation energy without a catalyst (E_{uncat}).
 - The activation energy with a catalyst (E_{cat}), which is lower.

Users must drag and drop the correct labels into the appropriate blank zones on the graph, including:

- The X-axis and Y-axis
- The red and blue curves
- The two activation energy lines near the curve tails

The drag-and-drop options are as follows:

T_2 higher temperature

T_1 lower temperature

Uncatalysed activation energy E_{uncat}

Catalysed activation energy E_{cat}

Energy

Number of particles

This interactivity helps users understand how temperature and catalysts affect the distribution of particle energies.

Read below for the solution:

Y-axis - Number of particles

X-axis - Energy

Red curve - T_1 lower temperature

Blue curve - T_2 higher temperature

First section - Catalysed activation energy E_{cat}

Second section - Uncatalysed activation energy E_{uncat}

This activity reinforces that: At **higher temperatures**, a **greater proportion of particles** have enough energy to overcome the activation barrier.

Catalysts lower the activation energy, thereby **increasing the number of particles** that can react.

Activity

- **IB learner profile attribute:**
 - Knowledgeable
 - Inquirers
 - Thinkers
 - Reflective
- **Approaches to learning:**
 - Thinking skills — Providing a reasoned argument to support conclusions; Applying key ideas and facts in new situations
 - Communication skills — Use terminology, symbols and communication conventions consistently and correctly
 - Social skills — Working collaboratively to achieve common goals; Appreciating the talents and diverse needs of others
- **Time required to complete activity:** 25 minutes
- **Activity type:** Individual/pair activity

1. A student wrote the following explanation for how a catalyst works:

‘Sulfur dioxide and oxygen react to give sulfur trioxide, in the presence of a vanadium pentoxide catalyst. For the reaction to occur the sulfur dioxide and oxygen molecules must collide. The catalyst lowers the activation energy of this collision.’

Identify the errors and suggest a better answer.

2. Define the term activation energy of a reaction.
3. (a) Draw, with labelled axes, the Maxwell-Boltzmann curve to show the distribution of molecular energies in a gas. Label this curve T_1 . Draw a second curve T_2 to show the same sample of gas at a higher temperature. (b) Draw two vertical lines showing the activation energy on the Maxwell-Boltzmann curve for the uncatalysed, E_a (uncat), and catalysed, E_a (cat) reactions. (c) Explain why altering the temperature alters the frequency of successful collisions. (d) What does the area under the curve T_1 represent and how is it different from the area under the curve T_2 .
4. Mark the activation energy of the backward reaction on the following energy profile diagram (**Figure 10**). Sketch what the curve would look like in the presence of a catalyst.

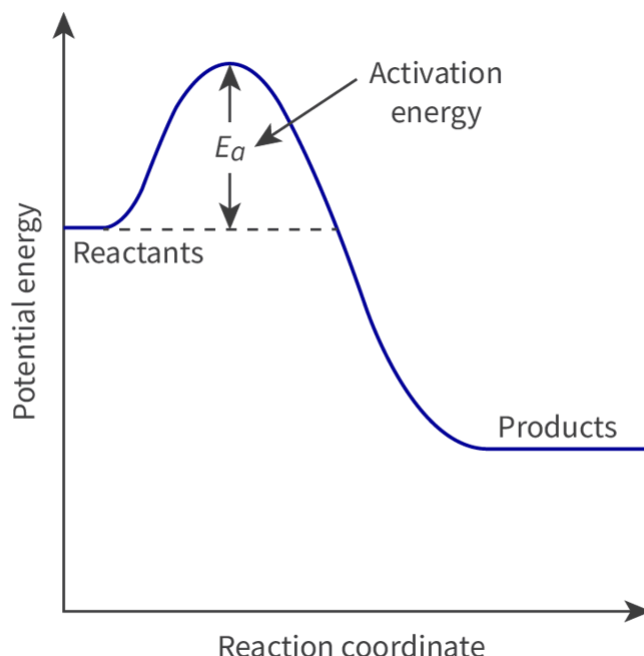


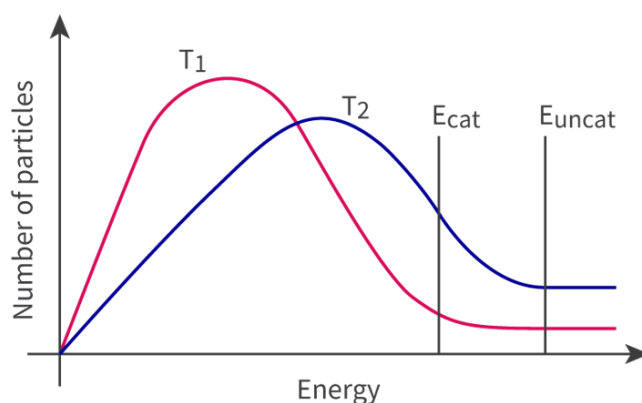
Figure 10. Energy profile diagram.

 More information for figure 10

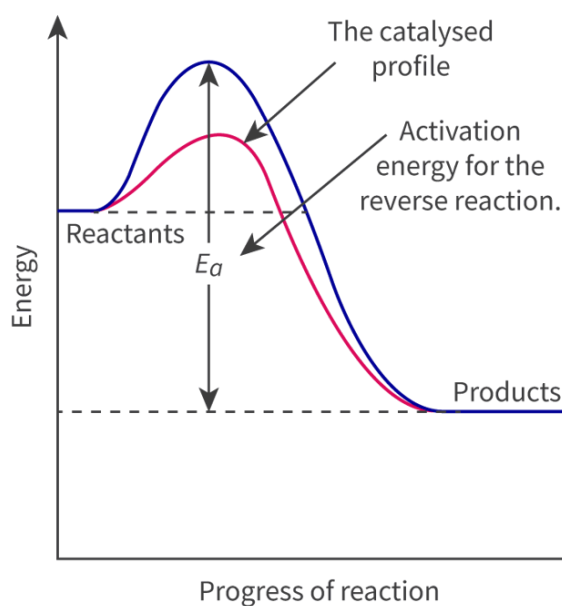
The diagram is an energy profile graph showing the relationship between potential energy and reaction coordinate. The X-axis represents the reaction coordinate, and the Y-axis represents potential energy. The graph features a curve that starts at a certain energy level, increases to a peak, and then decreases. This increase represents the activation energy (labeled as ' E_a '), which is the energy barrier that must be overcome for the reaction to progress from reactants to products. The reactants are labeled at the initial low-energy level prior to the increase, while the products are labeled at the lower energy level after the peak. Arrows indicate the direction from reactants to products along the reaction coordinate, and a dotted line depicts the energy level of reactants beneath the peak, highlighting the activation energy required for the reaction. The overall shape of the graph indicates an exothermic reaction, with products being lower in energy than reactants.

[Generated by AI]

1. The student identifies that the reaction is catalysed by vanadium pentoxide but it is not about the collisions between the molecules. The catalyst does not lower the activation energy but provides an alternative route through which the reaction can proceed. It is this route that has a lower activation energy.
2. The activation energy, E_a , is the minimum kinetic energy that the colliding molecules must possess for the collisions to be successful and result in the formation of a product molecule.
3. (a) and (b)



1. (c) Temperature is a measure of the average kinetic energy of the molecules. Increasing the temperature, for example, leads to an increase in kinetic energy, this would lead to an increase in the frequency of collisions.
 (d) The area under curve T_1 represents the total number of particles in the sample. The shape of the curve changes in T_2 , but the area under the curve still remains the same, as long as the number of particles is not altered.
2. The catalysed profile and activation energy of the reverse reaction:





Reaction mechanisms (HL)

Higher level (HL)

Learning outcomes

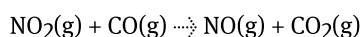
By the end of this section you should be able to:

- Evaluate proposed reaction mechanisms and recognise reaction intermediates.
- Distinguish between intermediates and transition states and recognise both in energy profiles of reactions.

In [subtopic S1.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/>) you learned how carrying out a chemical reaction is very similar to following a recipe and how the list of measured ingredients is similar to the stoichiometric quantities. The method for carrying out the recipe usually follows the ingredients and is specific for the recipe. Just like we would not dump in everything all at once in order to make cake, a reaction also proceeds in a sequence of steps, known as the reaction mechanism. The reaction mechanism is to a reaction what the instructions are to a recipe.

Individual steps in the mechanism are called elementary steps. For example, we might find out which bonds are broken first, when the transition state is formed and whether the reaction takes place in one or more elementary steps.

Reaction mechanisms are not something we can observe directly, they are based on hypotheses formulated using kinetic and stoichiometric data. The task of the chemist is to carry out experiments to propose a mechanism that best supports the experimental observations, in other words the proposed mechanism is a theoretical mechanism that is consistent with the experimental data. In fact, it is also possible to think of several mechanisms for a single reaction. Working out a mechanism for a reaction can be very challenging and fascinating at the same time. For example, in the reaction:



it has been shown that the reaction has the following possible mechanism:

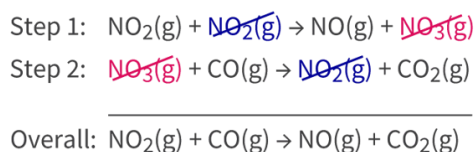


Figure 1. A reaction mechanism.

 More information for figure 1

The image illustrates a reaction mechanism consisting of two steps. Step 1 shows the reaction of NO₂ gas with another NO₂ molecule, producing NO gas and NO₃ gas, with the NO₃ being crossed out. Step 2 shows the reaction of NO₃ gas with CO gas, producing NO₂ gas and CO₂ gas, with the NO₂ being crossed out. The overall reaction, shown at the bottom, simplifies to NO₂ gas reacting with CO gas to produce NO gas and CO₂ gas. Cancelled species NO₃ and NO₂ are shown crossed out in the intermediate steps, indicating they are intermediates and are not present in the overall reaction.

[Generated by AI]

The overall reaction is a sum of the elementary steps and must be identical to the balanced equation. The overall reaction is obtained by adding the elementary steps together and cancelling species that exist on both sides of the reactions.

Notice that that one molecule, NO₃, appears in both steps but does not appear in the overall reaction (**Figure 1**). In this case NO₃ is a reaction intermediate. A reaction intermediate is a temporary molecule produced in one step and subsequently consumed in the next. An intermediate is typically unstable, so they may not be represented by formulas that are familiar to you.

The rate determining step

The rate determining step can be compared to a series of funnels (**Figure 2**). It is the funnel with the smallest diameter that controls how fast the liquid flows through, whether it is placed first or last. In the same way, in a chemical reaction occurring in several steps, it is the slowest step that determines the rate of the reaction.



Figure 2. The rate determining step can be compared to a series of funnels.

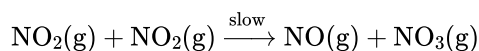
 More information for figure 2

The image is an illustration depicting three vertical arrangements of funnels with a blue liquid being poured into the top funnel of each set. Each column has a series of funnels stacked vertically. In each set, the diameter of one of the funnels is notably smaller, illustrating the concept of the rate-determining step. The flow of liquid is continuous from the top to the bottom funnel, eventually collecting in a flask at the base of each arrangement. This visual is used to explain how the narrowest point (smallest funnel) controls the overall rate of flow, similar to how the slowest step dictates the rate of a multi-step chemical reaction.

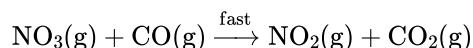
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For example in the reaction between nitrogen dioxide and carbon monoxide the rate determining step is the first step.

Step 1:



Step 2:



Step 1 limits the rate of reaction, and thus will determine the rate.

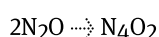
International Mindedness

Plastic pollution has become one of the major environmental issues threatening the planet today. The development of biodegradable plastics has been a positive step towards more sustainable solutions. Biodegradable plastics, as the name suggests, breakdown and recycle the elements they contain back into the soil or aquatic environment. Research into plastics involves studying the reaction mechanisms involved in the production and degradation process. The reaction mechanism gives details of how different atoms are reassembled during the process. These reactions can then be manipulated by changing reaction conditions and shuffling atoms, to obtain molecules with desirable properties, like easy decomposition under different conditions, strength and durability.

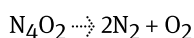
Worked example 1

A proposed reaction mechanism has the following steps:

Step 1 Fast:



Step 2 Slow:

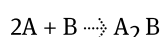


1. Write the overall equation.
 2. Which step is the rate determining step?
 3. Identify the intermediate.
-
1. Combine the equations and cancel the intermediate

$$2\text{N}_2\text{O} \rightarrow 2\text{N}_2 + \text{O}_2$$
 2. Step 2, it states it is Slow whereas Step 1 states it is Fast.
 3. Intermediate N_4O_2

Worked example 2

Propose two possible reaction mechanisms for the given reaction.



Step 1: $A + B \rightarrow AB$

Step 2: $AB + A \rightarrow A_2B$

Overall: $2A + B \rightarrow A_2B$

Notice AB is an intermediate in this reaction, which is produced in the first step and consumed in the next.

or

Step1: $A + A \rightarrow A_2$

Step 2: $A_2 + B \rightarrow A_2B$

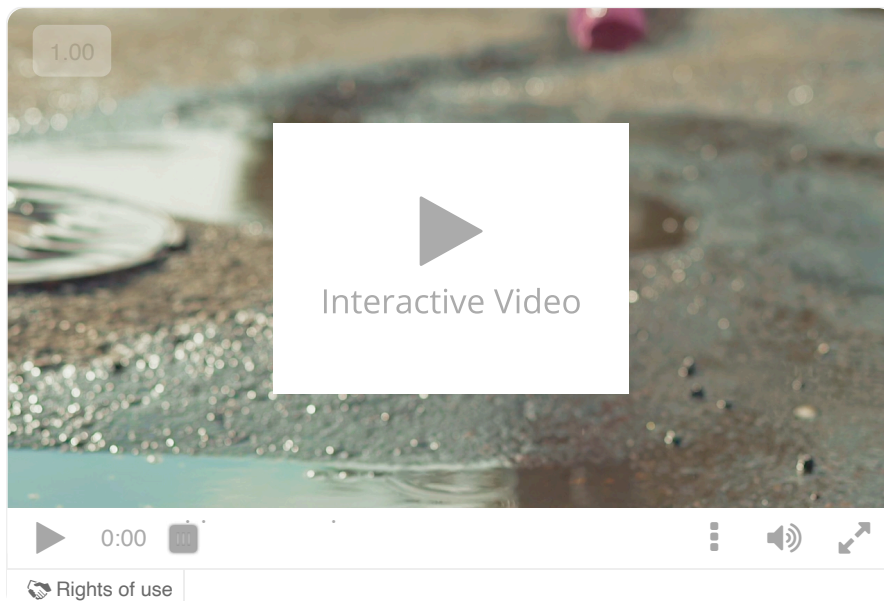
Overall: $2A + B \rightarrow A_2B$

A_2 here is an intermediate.

Transition states and reaction intermediates

In section R2.2.4–5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/maxwell-boltzmann-distribution-curves-id-44782/>) you learned about transition states. It is important to distinguish between transition states and intermediates in order to draw reaction mechanisms.

Imagine a walk through a park (**Interactive 1**), if you decide to jump over a series of puddles along a path, when you are in the middle of your jump, you are neither walking nor touching the ground, this is your transition state. You cannot freeze mid-air as you hover over a puddle at your transition state. The same thing applies to transition states in reactions, they are unstable and quickly revert into products. If then, upon landing you pause momentarily on one leg before jumping over the next puddle, this is your intermediate stage before you actually land on two feet and start walking again. You can stop at your intermediate, you might wiggle a little but are generally quite stable. A reaction intermediate is the same. It is relatively stable and in some cases stable enough to be isolated.



Interactive 1. Jumping Over Puddles.

 More information for interactive 1

An interactive video displays a child wearing rain boots and jumping over puddles to demonstrate the transition state and intermediate state of a reaction.

As the video progresses, it freezes when the child is in mid-air above the first puddle and a text appears as "the transition state". When the child lands in the puddle, this stage is referred to as "the intermediate state".

As the video continues further, the child takes two steps forward and jumps again. The video freezes mid-air to indicate another transition state. Then the child lands on the second puddle, causing the water to splash onto their legs and outside the puddle.

Transition states

The transition state and reaction intermediates can be represented in energy profile diagrams. For a single step reaction the transition state can be represented as shown in **Figure 3**. The transition state occupies the highest point on the reaction profile diagram, shown as the 'peak' of the hill.

- The transition state is not the reactant or product. It resembles both as only some bonds in the reactant are broken and some other bonds are formed.
- The transition state exists for a very short period of time and hence cannot be isolated from the reaction mixture.

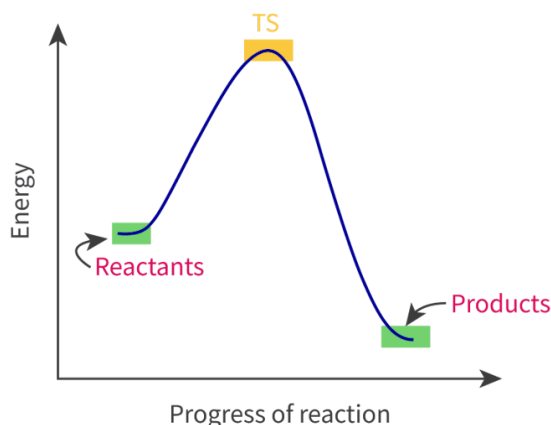


Figure 3. The energy profile diagram showing the transition state.

 More information for figure 3

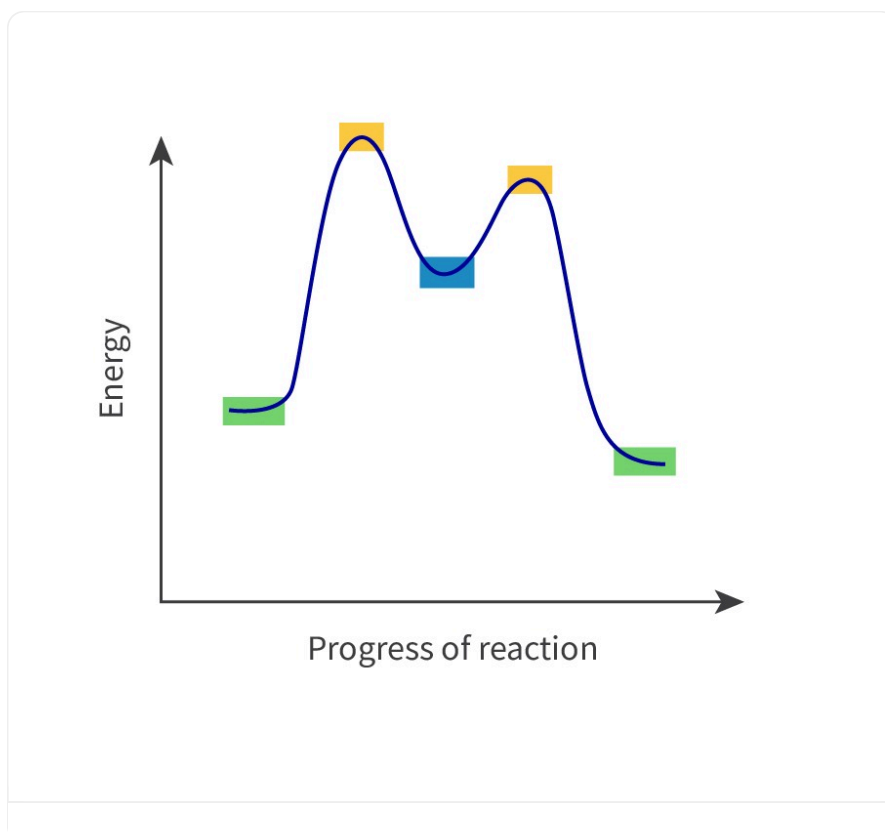
The image is a graph illustrating an energy profile diagram for a chemical reaction. The X-axis represents the 'Progress of reaction' while the Y-axis represents 'Energy.' The diagram features a curve that starts at the energy level of the reactants, rises to a peak indicating the transition state (TS), and then declines to the energy level of the products. The curve begins with reactants labeled at a lower energy point, ascends to the top where the transition state is labeled 'TS,' and descends to a lower energy point labeled 'Products.' The transition state marks the highest energy level on the graph. Arrow indicators show the progression from reactants to products, emphasizing the peak as a critical point in the reaction process.

[Generated by AI]

Intermediates

Intermediates are species with lower energy when compared to transition states. However, they are still quite unstable and in most reactions they cannot be isolated. Not all reactions have intermediates. There are plenty of single-step reactions that do not have intermediates.

Interactive 2 shows a two-step reaction. Can you identify where the transition states and intermediates would be on the energy profile diagram? Use the slider along the bottom of the energy profile to reveal the answer.



Interactive 2. The Energy Profile Diagram Showing the Energy of the Intermediate and Transition States.

 More information for interactive 2

This interactive slide graph illustrates the energy changes of reactants, intermediates, and products, along with their energy transition states throughout chemical reactions. The slider starts with a graph with the x-axis labeled as the progress of the reaction and the y-axis labeled as energy. Initially, the curve begins at a low energy point. As the curve rises, it reaches its first peak and after reaching the first peak, the curve decreases in energy, forming a low valley. It then rises again, forming a second peak. This peak again decreases, leading to a very low point. As the slider progress curve of the graph shows labelled positions. The first low energy point is labelled as reactants on the left side of the curve. The curve's first peak is labelled as TS1, which is the first transition state. The curve decreasing in energy is labelled as intermediate. The second peak of the rising curve is labelled as TS2 that is the second transition state. Finally, the curve ends with a low-energy point and is marked as Products. This interactive demonstrates how energy changes during a two-step reaction and explains key concepts such as transition states, intermediates, and overall reaction energy flow.

Table 1 shows a comparison between transition states and intermediates.

Table 1. Summary of the differences between transition states and intermediates.

Transition states	Intermediates
High energy state	Lies between two high energy stat
Have a very short lifetime so are unstable and cannot be isolated	Are usually unstable and in most reac cannot be isolated*
Every reaction has a transition state	Exist only in two or more step reacti
Some reactant bonds are broken and some product bonds are formed	All of the reactant bonds are broken to the intermediate product

*In general, reaction intermediates cannot isolated. However, there may be certain reactions where it is possible.

Activity

- **IB learner profile attribute:**
 - Knowledgeable
 - Inquirer
 - Thinker
 - Reflective
- **Approaches to learning:**
 - Thinking skills — Providing a reasoned argument to support a conclusion, Applying key ideas and facts in new contexts
 - Communication skills — Using terminologies, symbols and communication conventions consistently and correctly
 - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 30 minutes
- **Activity type:** Group activity

In this activity you will work in groups of five to ten.

Materials

- Two sets of ten unlabelled ping pong balls
- Two spoons or spatulas

Activity

This activity is a relay race between two groups of students.

- Divide the class into two groups and sit each group in a line with a basket of ten ping pong balls at one end of each line. The race will involve passing the balls from one basket to another empty basket at the other end. The team that passes all ten balls from one basket to the other first, wins.
- Most of you will use your hands. One person in each team must use a spoon or spatula and one more person in each team should stand up when he receives the ball, jump twice and turn in a circle before he passes the ball on.

Rule

If the person you are trying to pass the ball to already has a ball, hold on to yours. A person can only have one ball at a time.

In your teams discuss the following questions:

1. What was the main factor that influenced how fast the balls were passed?
2. Based on what you have learned from this activity, explain how this can be applied to reaction mechanisms and the rate of reaction.

Energy profiles and molecularity (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R2.2.7) Construct and interpret energy profiles from kinetic data.
- (R2.2.8) Interpret the terms unimolecular, bimolecular and termolecular.

Energy profiles from kinetic data

Imagine yourself trekking up and down a scenic hilly area. Connecting with nature in this way can be very rewarding. Along the way you will experience routes that may offer considerable resistance and will require more input from you. Other times you may literally be running downhill.

Figure 1 maps the peaks and valleys for the Inca Trail.

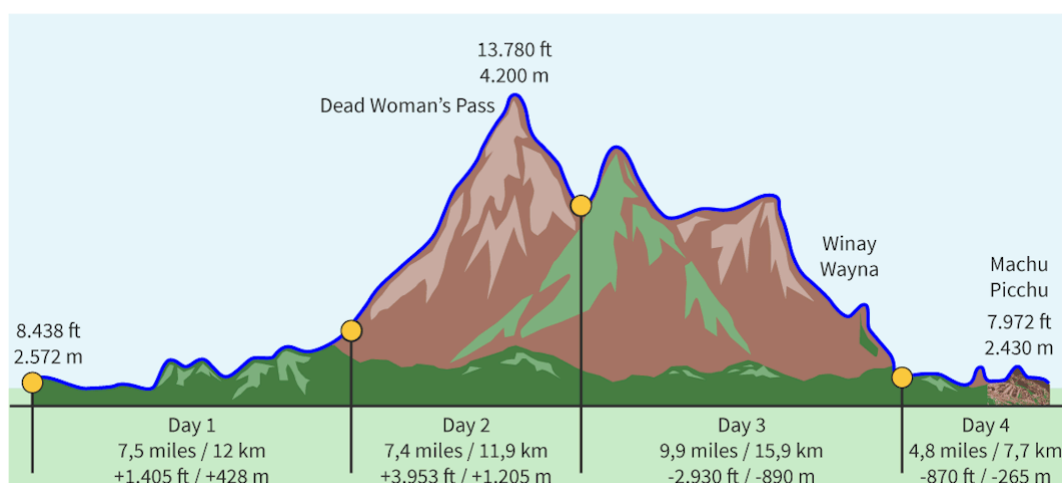


Figure 1. The Inca Trail map to Machu Picchu, in Peru, South America.

 More information for figure 1

The image is an elevation profile of the Inca Trail to Machu Picchu. It shows a mountainous landscape divided into four segments, each representing a day of the hike.

- **Day 1:** Starts at 8,438 feet (2,572 meters) with a distance of 7.5 miles (12 km) and an elevation gain of 1,405 feet (428 meters).
- **Day 2:** The trail ascends sharply to Dead Woman's Pass, the highest point at 13,780 feet (4,200 meters). This segment covers 7.4 miles (11.9 km), gaining 3,953 feet (1,205 meters) in elevation.
- **Day 3:** The trail descends to Winay Wayna, covering 9.9 miles (15.9 km) and losing 2,930 feet (890 meters) in elevation.
- **Day 4:** Ends at Machu Picchu at 7,972 feet (2,430 meters), with a distance of 4.8 miles (7.7 km) and an elevation loss of 870 feet (265 meters).

Key locations such as Dead Woman's Pass and Winay Wayna are marked along the top of the profile. The blue line traces the trail over the mountains depicted in brown and green hues.

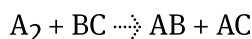
[Generated by AI]

Similarly, the path that reactions take can be represented to show the different steps and their related energy in an energy profile diagram. In [section R2.2.6 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reaction-mechanisms-hl-id-45509/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reaction-mechanisms-hl-id-45509/) you learned how transition states and intermediates could be represented on energy profile diagrams. Energy profile diagrams can be used to show multi step reactions and can give useful information on the mechanism of the reaction. A multi-step reaction profile will be composed of a whole 'mountain range' with a series of dips and peaks, just like our Inca trail above. Take for example this reaction:

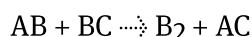


The proposed mechanism for this reaction is:

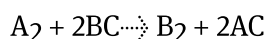
Step 1:



Step 2:



Overall:



This reaction can be represented on an energy profile diagram as shown in **Figure 2**.

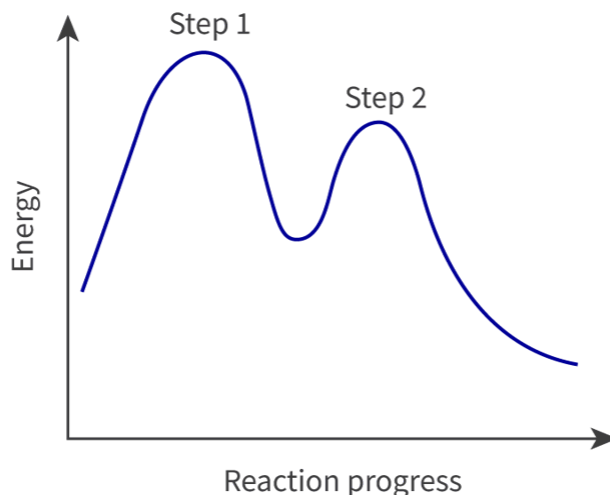


Figure 2. An energy profile for a multi-step reaction.

 More information for figure 2

The image is a graph depicting an energy profile for a multi-step reaction. The X-axis is labeled 'Reaction progress,' and the Y-axis is labeled 'Energy.' The graph features a blue line that forms two peaks. The first peak is higher and labeled 'Step 1,' indicating an initial increase in energy. Following Step 1, the energy decreases before rising again to a second, smaller peak labeled 'Step 2.' After Step 2, the energy decreases continuously, suggesting the progression of the reaction. This graph provides insight into which step may be considered the rate-determining step based on the height of the peaks.

[Generated by AI]

Just by looking at the energy profile diagram in **Figure 2** can you tell which step is the rate determining step?

What information can be deduced from the energy profile diagram?

- We can see that it is a two-step reaction because there are two 'peaks' representing the transition states for each elementary step and one 'dip' representing the intermediate.
- The first transition state is at a higher energy level than the second, so the activation energy for the first step is higher than that for the second step. This means the first step is the slow step, or rate-determining step.
- The enthalpy change of the reaction is the distance between the reactants and products and can be marked on the diagram as shown in **Figure 3**.

The activation energies for the first step and second steps are E_{a1} and E_{a2} respectively. ΔH represents the enthalpy change of the reaction.

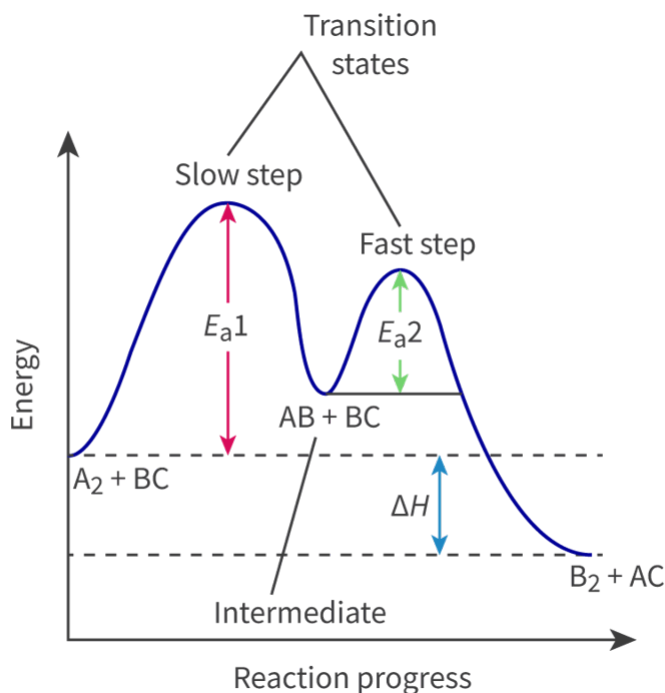


Figure 3. An energy profile diagram for a multi-step reaction showing the activation energies, E_{a1} and E_{a2} and the enthalpy change (ΔH) of the reaction.

 More information for figure 3

The image displays an energy profile diagram for a multi-step chemical reaction. The X-axis represents 'Reaction progress' and spans from 'A2 + BC' to 'B2 + AC'. The Y-axis represents 'Energy'. Two peaks are evident, indicating transition states labeled as 'Slow step' and 'Fast step'. The first peak, representing the slow step, has a higher activation energy labeled ' E_{a1} '. The second, smaller peak represents the fast step with activation energy ' E_{a2} '. The intermediate stage is depicted between these peaks. A downward arrow labeled ' ΔH ' indicates the enthalpy change from the top of the fast step to the product level. Key labels include 'AB + BC' within the intermediate area. The position of each transition state is marked with corresponding arrows indicating the direction of energy changes needed to proceed through each step.

[Generated by AI]

The overall activation energy of a multi-step reaction

It is very important to understand that a multi-step reaction may have different activation energies at each step, however the minimum energy required for the reaction to proceed must be equal to the energy needed to take the reactants to the transition state with the highest energy. **Figure 4** illustrates the activation energy for a multi-step reaction.

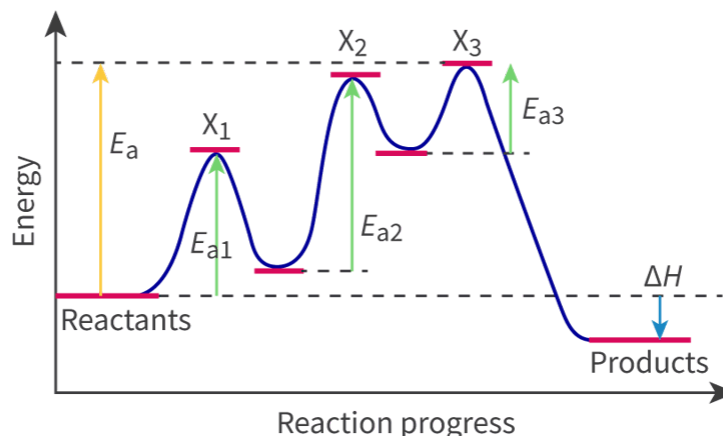


Figure 4. An energy profile diagram showing the activation energy of a multi-step reaction.

 More information for figure 4

The image is a diagram illustrating the energy profile of a multi-step chemical reaction. The X-axis represents the reaction progress, ranging from Reactants on the left to Products on the right. The Y-axis represents the energy, with labels indicating different energy levels throughout the reaction.

There are three distinct energy peaks corresponding to three transition states labeled X1, X2, and X3. Each peak has an associated activation energy, labeled E_{a1} , E_{a2} , and E_{a3} , respectively. The overall activation energy, E_a , is represented by a yellow arrow pointing from the base energy level of the reactants to the top of the highest peak, X3. This step is identified as having the highest energy and is the rate-determining step.

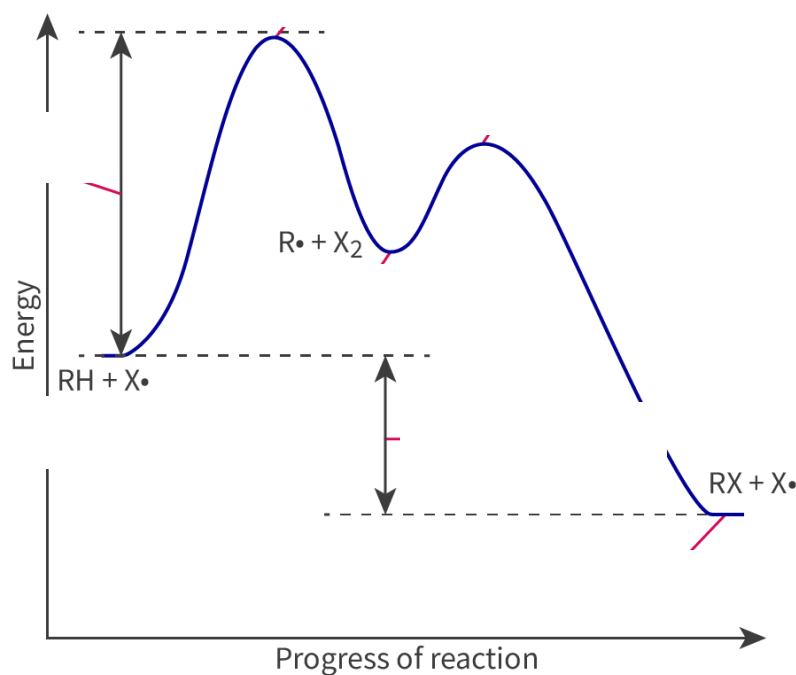
The diagram shows energy valleys between each peak, representing intermediate steps in the reaction. The change in enthalpy, ΔH , is depicted by a blue arrow pointing downward from the reactants to the products.

The different components are interconnected by a blue curve outlining the energy path of the reaction, illustrating the energy changes at each step.

[Generated by AI]

The overall activation energy E_a is represented by the yellow arrow going from the reactants to the top of the third step which has the transition state at the highest energy X3. The rate determining step is the second step with the highest activation energy E_{a2} .

You can practise labelling energy profile diagrams by putting the following labels into their correct place by using **Interactive 1**.



Fast step

Slow step

Enthalpy change

Starting materials

Intermediates

Activation energy

Products

☒ Check

Interactive 1. Energy profile diagrams.

More information for interactive 1

An interactive energy profile diagram plots energy versus the progress of a reaction. A curve starts with the label $RH + X\cdot$, increases to form a large first peak, dips into a valley labelled $R\cdot + X_2$, increases again to form a small second peak, and finally descends to end at the label $RX + X\cdot$. Users need to drag and drop the correct terminologies into the corresponding missing drop zones, which include:

 $RH + X\cdot$

First peak

 $R\cdot + X_2$

Second peak

 $RX + X\cdot$

Distance between $\text{RH} + \text{X}\cdot$ and the first peak

Distance between $\text{RH} + \text{X}\cdot$ and $\text{RX} + \text{X}\cdot$

The drag-and-drop options provided are:

Fast step

Slow step

Enthalpy change

Starting materials

Intermediates

Activation energy

Products

This interactivity helps users to interpret an energy profile diagram for a multi-step chemical reaction and distinguish between fast and slow steps based on activation energy peaks.

Read below for the solution:

$\text{RH} + \text{X}\cdot$ - Starting materials

First peak - Slow step

$\text{R}\cdot + \text{X}_2$ - Intermediates

Second peak - Fast step

$\text{RX} + \text{X}\cdot$ - Products

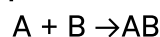
Distance between $\text{RH} + \text{X}\cdot$ and the first peak - Activation energy

Distance between $\text{RH} + \text{X}\cdot$ and $\text{RX} + \text{X}\cdot$ - Enthalpy change

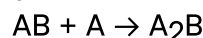
Worked example 1

Sketch the energy profile diagrams for the following reaction mechanisms. Label any intermediates on the energy profile diagram,

1. Step 1, Fast:

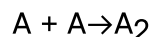


Step 2, Slow:

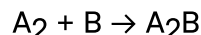


Given that this reaction is endothermic.

2. Step 1, Slow:

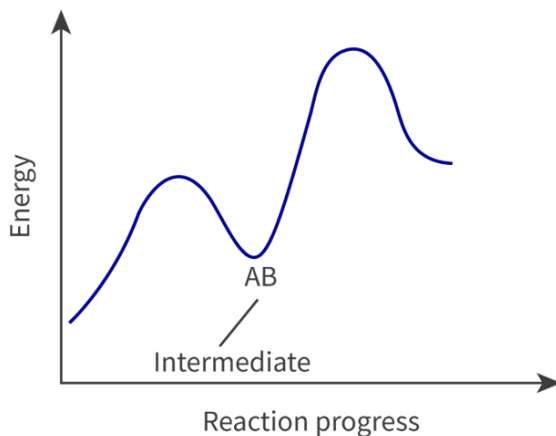


Step 2, Fast:

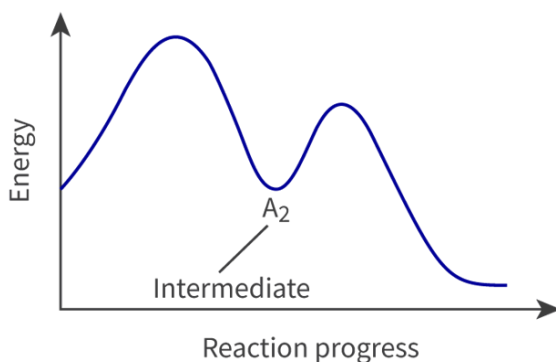


Given that this reaction is exothermic.

1.



1.



Molecularity of a reaction

For any elementary step in a reaction mechanism it is very important to know the number of reactant species involved. The term molecularity refers to the number of species reacting in an elementary step.

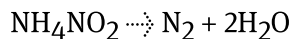
An elementary step where there is only one reactant is referred to as a unimolecular step or is said to have a molecularity of one.

Bimolecular refers to an elementary step with two reactant particles and the elementary step is said to have a molecularity of two.

A termolecular step, having a molecularity of three, is very rare. Why do you think this is so? It is because the probability of three reactant particles colliding with sufficient energy and in the correct orientation is very low.

Molecularity is always a whole number and is never greater than three.

For example, the following reaction is unimolecular, as there is only one reactant molecule (as seen directly from the stoichiometry of the reaction):

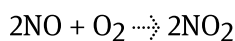


in contrast, this reaction:



is a bimolecular reaction as two HI molecules are colliding with the proper orientation and sufficient energy.

The following example is that of a termolecular reaction:



1.00

Unimolecular reaction Bimolecular reaction Termolecular reaction

0:00

Rights of use

Interactive 2. Molecularity.

More information for interactive 2

A video depicts three different reactions labelled unimolecular, bimolecular, and termolecular, demonstrating how products are formed in each case.

In the unimolecular reaction, a single reactant labelled A is converted into a product.

In the bimolecular reaction, two reactants, labelled A and B, react together to form a product.

In the termolecular reaction, three reactants, labelled A, B, and C, react together to form a product.

This video provides a comparison of unimolecular, bimolecular, and termolecular reactions, highlighting their differences in reactant participation and product formation.

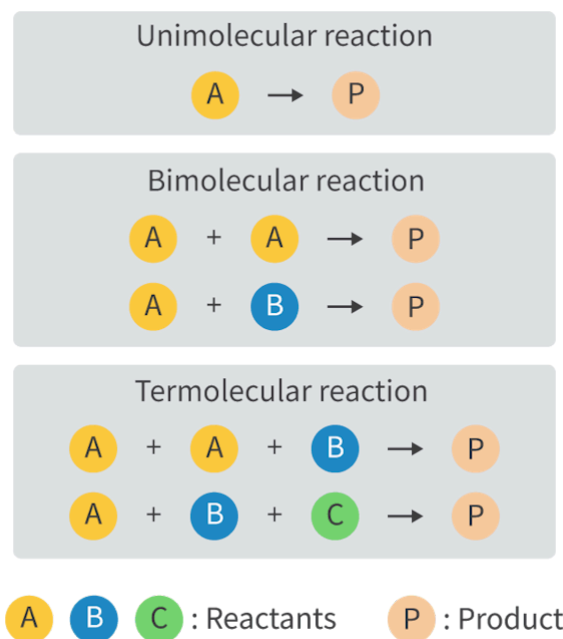


Figure 5. Molecularity in reactions.

 More information for figure 5

The diagram illustrates different types of chemical reactions: unimolecular, bimolecular, and termolecular.

1. **Unimolecular Reaction:** This section shows a single reactant 'A' transforming into a product 'P'.
2. **Bimolecular Reaction:** This section contains two subsections:
 3. The first shows two 'A' reactants combining to form a product 'P'.
 4. The second shows one 'A' and one 'B' reactant combining to form a product 'P'.
5. **Termolecular Reaction:** This part includes two subsections:
 6. The first shows two 'A' reactants and one 'B' reactant combining to form a product 'P'.
 7. The second shows one 'A', one 'B', and one 'C' reactant combining to form a product 'P'.

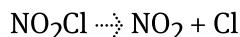
Below the reactions, a legend explains the color-coded symbols: 'A', 'B', and 'C' are reactants, while 'P' is the resulting product. The diagram uses arrows to indicate the direction of each reaction.

[Generated by AI]

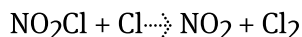
Worked example 2

One of the proposed mechanisms for the thermal decomposition of nitryl chloride is:

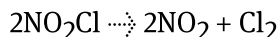
Step 1 (slow step):



Step 2 (fast step):



Overall:



Identify the molecularity of the elementary steps in this mechanism.

Step 1 is unimolecular

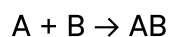
Step 2 is bimolecular

Activity

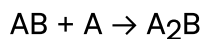
- **IB learner profile attribute:**
 - Knowledgeable
 - Inquirer
 - Thinker
 - Reflective
- **Approaches to learning:**
 - Thinking skills — Providing a reasoned argument to support a conclusion, Applying key ideas and facts in new contexts
 - Communication skills — Using terminologies, symbols and communication conventions consistently and correctly
 - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 15—20 minutes
- **Activity type:** Group activity

Work in groups of three to make an energy profile diagram for the following two-step exothermic reaction:

Step 1:



Step 2:



Each person in the group will make a separate energy profile diagram. One showing the first step as the slow step and one where the second step is the slow step.

1. Compare the profile diagrams in each group and deduce the overall equation.
2. Identify the molecularity of each elementary step.

The rate expression and its experimental determination (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to deduce the rate equation for a reaction from experimental data.

Some of us use route planner apps to plan out a journey. This tells us how long it will take to get to our destination for various different routes. It also tells us how long it will take if we use a different mode of transport. How does it do that at the touch of a button? Most of these apps work on some kind of mathematical model or equation which is solved by putting in values for different variables, like distance (often calculated automatically from start and end point) and mode of transport. The benefit of equations is that they provide a very precise way to deduce the impact of changing variables on various phenomena in the world.

Google maps, for example, is an extended app that is also capable of accounting for any congestion on the way to give you an estimated arrival time. This means it uses an expanded model that takes into account many other variables like bottlenecks or incidents that may occur across the route.

A rate expression or rate law does exactly that. It is a mathematical equation which calculates the effect of change in concentration of various reactants on the rate of reaction.

In section R2.2.2–3a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/factors-affecting-the-rate-of-reaction-id-44714/>) you learned how a change in concentration of the reactants changes the rate of the equation. A rate equation shows this effect mathematically.

Measuring the rate of reaction

For the purposes of rate equations, the rate of a reaction is measured in terms of how fast the concentration of one of the reactants is falling. Its units are $\text{mol dm}^{-3} \text{s}^{-1}$.

We know that increasing the concentration of the reactants increases the overall rate of the reaction, but not all reactants affect the rate in the same way. Some reactants have a greater influence on the rate of reaction than others. Why do you think that is? Think about reaction mechanisms and rate determining steps. Do all the reactants impact the rate of reaction similarly? In order to understand how reactants can influence the rate of a reaction we need to explore further.

Suppose you carry out an experiment to see how the rate of reaction changes when you alter the concentration of a reactant, A, Some of the possibilities are:

I. The rate of reaction is directly proportional to the concentration of A

This means if you double the concentration of A the rate also doubles. Increasing the concentration three times increases the rate by three times. We can write this as:

$$\text{rate} \propto [\text{A}]$$

II. The rate of reaction is proportional to the square of the concentration of A

If you were to double the concentration of A the rate will go up four times (2^2) or if you increase the concentration of A four times the rate goes up sixteen times ($4^2=16$). We can write this as:

$$\text{rate} \propto [\text{A}]^2$$

III. Extending the possibilities to other reactants

By performing similar experiments for reactant B, in a reaction between A and B, you would see that the reactant concentrations are related to the rate in the following way:

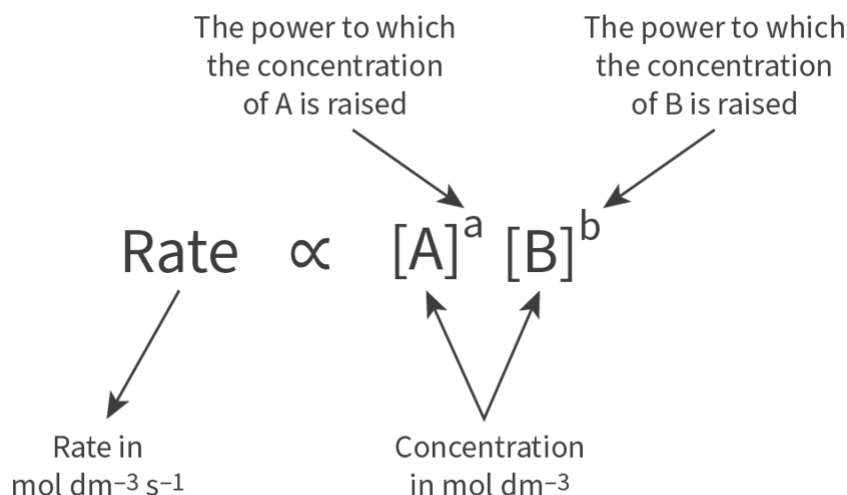


Figure 1. The rate expression for the reaction in which A and B are reacting.

 More information for figure 1

The image is a diagram explaining the rate expression for a chemical reaction involving two reactants, A and B. It shows the proportionality relation as "Rate $\propto [\text{A}]^a [\text{B}]^b$ ". This indicates that the rate is proportional to the concentrations

of A and B, raised to their respective powers, a and b. Arrows point out and describe key parts of the equation:

- An arrow pointing to "Rate" is labeled "Rate in mol dm⁻³ s⁻¹", specifying the units for the rate of reaction.
- The concentration of reactant A is labeled as "[A]" and has an accompanying arrow stating "The power to which the concentration of A is raised", indicating its exponent "a".
- Similarly, for reactant B, "[B]" is labeled with an arrow pointing to its exponent, "b", indicating "The power to which the concentration of B is raised".
- Another set of arrows points to both [A] and [B], specifying "Concentration in mol dm⁻³" for these reactants.

[Generated by AI]

If the concentration of a reactant has no effect on the rate, then the concentration of the reactant is raised to a zero. Anything raised to the power of zero is one, this means that the concentration of this reactant does not appear in the rate expression. Therefore, if doubling the concentration of A doubles the rate and changing the concentration of B has no effect on the rate of reaction, the rate expression can be written as:

$$\text{rate} \propto [\text{A}]$$

where the concentration of B is not included in the rate expression.

Moreover, if the rate expression for the reaction between A and B is:

$$\text{rate} \propto [\text{A}][\text{B}]$$

It means doubling the concentration of A will double the reaction rate. The same goes for B, doubling the concentration of B doubles the rate of reaction. That is the rate of reaction is directly proportional to the concentration of both A and B.

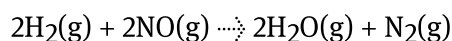
A reaction with the rate expression:

$$\text{rate} \propto [\text{A}]^2$$

means that doubling the concentration of A increases the rate four times. The concentration of B has no effect on the rate of reaction.

Worked example 1

In the reaction:



The rate of reaction doubles when the concentration of H₂ doubles.

Doubling the concentration of NO increases the rate four times.

Write the rate equation for this reaction.

$$\text{rate} \propto [\text{NO}]^2 [\text{H}_2]$$

Determining the rate expression experimentally using the initial rates method

Determining rate expressions experimentally involves determining the effect of changing concentration on rate with respect to each reactant individually. The initial rates method is used and as the name implies, the initial rates of the reactants are used to determine their effect on the rate.

Consider a reaction with two reactants A and B. By measuring the rate of reaction with different initial concentrations of A, while keeping the concentration of B constant, the effect of different concentrations of A can be found. For example, if changing the concentration of A has no effect on the rate it means the order of reaction with respect to A is zero. If the rate doubles when the concentration is doubled it means that the order of reaction is one with respect to A. If doubling the concentration of A results in a four times increase in rate it means the reaction is second order with respect to A.

These steps can then be repeated with different initial concentrations of B, while A is kept constant. This means any change in rate is only due to the concentration of B.

Now try the following worked example to see how initial rate data can be used to deduce the rate expression of a reaction.

Worked example 2

Use the data in the table to work out the rate expression for the reaction between A and B.

Experiment number	Concentration / mol dm^{-3} [A]	Concentration / mol dm^{-3} [B]
1	0.100	0.100
2	0.100	0.200
3	0.200	0.100

Looking at experiment 1 and experiment 2.

Doubling the concentration of B while keeping the concentration of A constant results in no change in the rate of reaction. This means that doubling the concentration of B has no effect on the rate of reaction. Thus the concentration of B will be raised to the power of zero and will not appear in the rate expression.

Looking at experiment 1 and experiment 3.

Doubling the concentration of A while keeping the concentration of B constant results in a four times increase in the rate of reaction. This means that doubling the concentration of A increases the rate four times (2^2). Thus the concentration of A will be raised to the power of two.

Hence the overall rate expression is:

$$\text{rate} \propto [\text{A}]^2$$

Now try a worked example that is a little bit more difficult.

Worked example 3

Use the data in the table to work out the rate expression for the reaction between A and B.

Experiment number	Concentration / mol dm^{-3} [A]	Concentration / mol dm^{-3} [B]
1	0.007	0.004
2	0.014	0.004
3	0.014	0.008

Looking at experiment 1 and experiment 2.

Doubling the concentration of A while keeping the concentration of B constant results in the rate increasing from $2.25 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$ to $9 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$.

In order to see how much rate has changed divide the initial rate of experiment 2 by the initial rate of experiment 1:

$$\frac{\text{initial rate experiment 2}}{\text{initial rate experiment 1}} = \frac{9 \times 10^{-4}}{2.25 \times 10^{-4}} = 4$$

This means doubling the concentration of A increases the rate four times. Thus the concentration of A will be raised to the power of 2 in the rate equation.

Looking at experiment 2 and experiment 3.

Doubling the concentration of B while keeping the concentration of A constant results in the rate increasing from $9 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$ to $1.8 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$.

In order to see how much rate has changed divide the initial rate of experiment 3 by the initial rate of experiment 2.

Going from experiment 2 to 3, the concentration of B has doubled while the concentration of A remains constant. In order to calculate the factor by which rate has changed:

$$\frac{\text{initial rate experiment 3}}{\text{initial rate experiment 2}} = \frac{1.8 \times 10^{-3}}{9 \times 10^{-4}} = 2$$

This means doubling the concentration of B doubles the rate, hence the rate of reaction is directly proportional to the concentration of B. Thus the concentration of B will be raised to the power of 1 in the rate expression.

Hence the overall rate expression is:

$$\text{rate} \propto [\text{A}]^2[\text{B}]$$

Table 1. A summary of how changing the concentration affects the rate of reaction when the concentration of the reactant is changed.

Changing the concentration of Reactant A	When the concentration of A is raised to the power of 1	When the concentration of A is raised to the power of 2
Doubling [A]	Doubles rate	Rate increases 4×
Tripling [A]	Triples rate	Rate increases 9×
Quadrupling [A]	Quadruples rate	Rate increases 16×

Worked example 4

The reaction $A + B \rightarrow A_2B$ has the rate expression of rate $\propto [A][B]^2$.

What will the effect on the rate of reaction be if the concentration of A is first tripled and then the concentration of B is halved?

$$\text{rate} \propto [A]$$

$$\text{let } [A] = a$$

then

$$\text{rate} \propto 3a$$

Tripling the concentration of A means the rate will increase three times.

$$\text{rate} \propto [B]^2$$

$$\text{Let } [B] = b$$

then

$$\text{rate} \propto \left(\frac{b}{2}\right)^2$$

$$\text{rate} \propto \frac{b^2}{4}$$

Halving the concentration of B means the rate will be reduced to one quarter of its original.

The net effect is the rate of reaction will increase by 0.75 times (3/4).

Activity

- **IB learner profile attribute:**
 - Knowledgeable
 - Inquirer
 - Thinker
 - Reflective
- **Approaches to learning:**
 - Thinking skills — Providing a reasoned argument to support a conclusion, Applying key ideas and facts in new contexts
 - Communication skills — Using terminologies, symbols and communication conventions consistently and correctly

- Social skills — Working collaboratively to achieve a common goal
- Time required to complete activity: 45 minutes
- Activity type: Group activity

In this activity you will work in three small groups to simulate the connection between reactant concentration and the rate equation.

Materials

- three small boxes of equal volume
- fifteen red spheres/ping pong balls
- fifteen blue spheres/ping pong balls

Preparation of boxes

Place five red and five blue spheres in box 1.

Place three red and seven blue spheres in box 2.

Place seven red and three blue spheres in box 3.

Alternatively you could carry out the activity by drawing diagrams of the boxes containing red (A) and blue (B) spheres.

Data provided

Each box represents the reaction between A and B. The red spheres represent A and the blue spheres represent B. Since all the boxes are of the same volume the number of spheres in each box represents their concentration.

The rate of reaction for each box is represented as:

$$\text{rate} \propto [\text{A}]^a [\text{B}]^b$$

The rates of reaction in each box are:

Box 1: rate $\propto 125$

Box 2: rate $\propto 147$

Box 3: rate $\propto 63$.

Task

1. Assign one box to each group.
2. In your individual groups, brainstorm using the data provided and the concentration of the reactants in your box to figure out the rate expression for the reaction between A and B.

3. Once everyone has come up with an expression, compare results. Is there any group that does not agree?
4. Remember all three boxes represent the same reaction between A and B. Is it possible to come to a mutual agreement about the rate expression?

The rate expression is $\text{rate} \propto [\text{A}] [\text{B}]^2$

Orders of reaction and reaction mechanism (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R2.2.10) Identify the number of particles in the rate determining step from the order with respect to a reactant and deduce the overall order of reaction.
- (R2.2.10) Sketch, identify and analyse graphical representations of zero order, first order and second order reactions.
- (R2.2.11) Solve problems using the rate equation including finding the units of k .

In chemical reactions, changing the concentration of a reactant typically results in a change in the rate. However, not all reactants have the same effect on the rate. Experimental evidence shows that some reactants have a greater effect on the rate than other reactants and there are situations where changing the concentration of one of the reactants has no effect on the rate at all – how can this be? How does your understanding of reaction mechanisms and molecularity support the experimental evidence of rates of reaction?

Orders of reaction and their graphical representation

For a reaction between A and B the powers to which the concentrations of the reactants are raised are called the orders of reaction with respect to A and B. The orders of reaction you are going to deduce are 0, 1 and 2. An order of 3 is also possible but rarely seen. For example, if the rate of reaction is directly proportional to the concentration of the reactant A (that is the concentration of A is raised to the power of 1) then the order of reaction with respect to A is 1. Similarly, if the rate of reaction is proportional to the square of the concentration of B then the order of reaction with respect to B is 2.

When the concentration of a reactant does not affect the rate, the order of reaction with respect to this reactant is 0 as the concentration of the reactant is raised to the power of zero. Since anything raised to the power of zero is 1, then the concentration of this reactant will not appear in the rate expression.

The overall order of the reaction is expressed by adding the individual orders of each reactant. So if the reaction is first order with respect to A, that is, $\text{rate} \propto [\text{A}]$, and second order with respect to B, that is, $\text{rate} \propto [\text{B}]^2$, then the overall order of reaction is three ($1 + 2 = 3$).

The third order rate equation can be expressed as:

$$\text{rate} \propto [\text{A}] [\text{B}]^2$$

introducing the proportionality constant gives:

$$\text{rate} = k [\text{A}] [\text{B}]^2$$

The proportionality constant is known as the rate constant, k . It is constant for each reaction given that the only possible changes are to the concentrations of the reactants. The effect of temperature, catalyst and activation energy on the rate constant will be discussed in [section R2.2.12–13](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/arrhenius-equation-hl-id-45513/>).

Study skills

The rate constant is denoted by the small, italicised letter k .

You can use a concentration–time graph, see [section R2.2.1](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-rate-of-reaction-id-43481/>), to see how the concentration of any reactant changes over time and how it affects the rate of the reaction. However, it is very difficult to differentiate between first order and second order graphs. A better picture of the effect of concentration on the rate can be obtained using rate-concentration graphs.

Zero order reaction

Changing the concentration of a reactant A does not affect the rate of reaction. The reaction is said to be zero order with respect to A.

A zero order reaction can be represented graphically as:

- Concentration–time graph: a straight line is obtained with a negative gradient.
- Rate–concentration graph: a horizontal line is produced representing a constant rate.

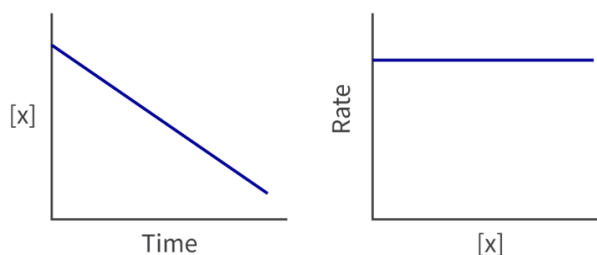


Figure 1. A concentration–time graph and a rate–concentration graph for a zero order reaction.

 More information for figure 1

The image displays two graphs side by side. The left graph is a concentration–time graph for a zero order reaction. The X-axis represents time, and the Y-axis represents concentration $[x]$. The line on this graph is a straight diagonal decreasing from the top left to the bottom right, indicating that concentration decreases linearly over time. The right graph is a rate–concentration graph for a zero order reaction. The X-axis represents concentration $[x]$, and the Y-axis represents the rate. This graph shows a horizontal line, indicating that the rate remains constant regardless of the concentration.

[Generated by AI]

First order reaction

When the concentration of a reactant A is raised to the power of 1, it means the concentration is directly proportional to the rate. The reaction is first order with respect to A and the rate equation can be written as $\text{rate} = k[A]$.

The overall order of reaction is also 1 in this case.

A first order reaction can be represented graphically as:

- Concentration—time graph: a curve showing a decreasing rate of reaction.
- Rate—concentration graph: a straight line passing through the origin with a positive gradient.

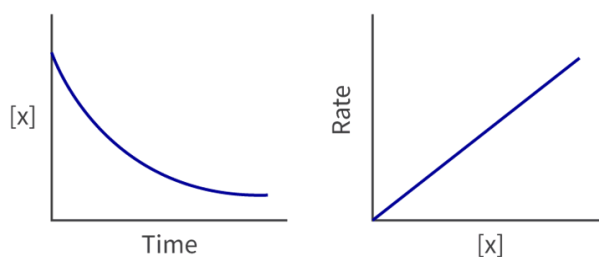


Figure 2. A concentration—time graph and a rate—concentration graph for a first order reaction.

 More information for figure 2

The image contains two separate graphs. The first graph on the left is a concentration-time graph. The X-axis represents time, with no specific values labeled, indicating a continuous progression. The Y-axis represents concentration, labeled as '[x]'. The graph shows a curve that decreases over time, depicting the diminishing concentration of a reactant in a first order reaction as time progresses.

The second graph on the right is a rate-concentration graph. The X-axis is labeled with '[x]', representing concentration, and the Y-axis is labeled 'Rate'. This graph illustrates a straight line with a positive slope, showing that the rate of reaction increases linearly with an increase in concentration in a first order reaction. Each axis is clearly labeled to represent the relationship between concentration and rate.

[Generated by AI]

Second order reactions

A reaction is said to be second order with respect to reactant A if the concentration of A is raised to the power of two. This means doubling the concentration of A increases the rate four times. The rate equation can be written as $\text{rate} = k[A]^2$.

A second order reaction can be represented graphically as:

- Concentration—time graph: as the concentration decreases the rate also decreases, in the form of a negative curve that slopes downwards.
- Rate—concentration graph: a curve is produced for this as well, but this curve is a positive curve, sloping upwards.

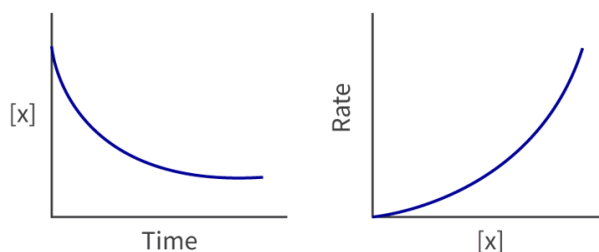


Figure 3. A concentration—time graph and a rate—concentration graph for a second order reaction.

 More information for figure 3

The image features two graphs. The first graph on the left is a concentration versus time graph for a second order reaction. The X-axis is labeled 'Time' and the Y-axis is labeled '[x]', representing concentration. The curve on this graph shows a downward slope, representing a decrease in concentration over time.

The second graph on the right is a rate versus concentration graph. The X-axis is labeled '[x]' indicating concentration, and the Y-axis is labeled 'Rate'. The curve on this graph slopes upward, indicating the rate increases with concentration.

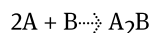
[Generated by AI]

Study skills

Keep in mind, the order of reaction can only be determined experimentally. You cannot deduce anything about the mechanism or order of reaction by looking at the reaction equation.

Worked example 1

A number of rate expressions have been given for the reaction



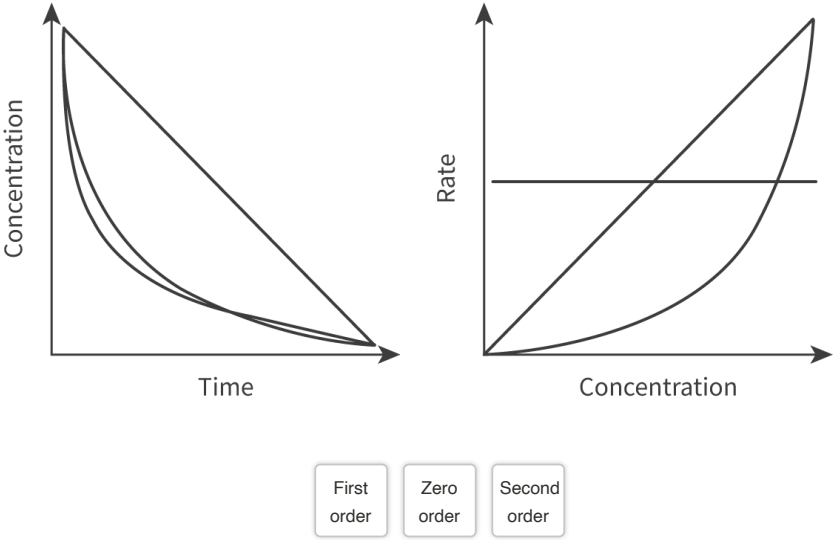
For each of the rate expressions identify the rate order with respect to each reactant and also identify the overall order of reaction.

1. rate = $k[A][B]$
2. rate = $k[A]^2[B]$
3. rate = $k[A][B]^2$
4. rate = $k[B]^2$

1. The reaction is first order with respect to A and first order with respect to B. The overall order of reaction is 2 ($1 + 1 = 2$).
2. The reaction is second order with respect to A and first order with respect to B. The overall order of reaction is 3 ($2 + 1 = 3$).
3. The reaction is first order with respect to A and second order with respect to B. The overall order of reaction is 3 ($1 + 2 = 3$).

4. The reaction is zero order with respect to A and second order with respect to B. The overall order of reaction is 2 ($0 + 2 = 2$).

Test your understanding of concentration–time graphs and rate–time graphs by dragging the correct labels onto the graphs in **Interactive 1**. Start by labelling the axes.



First order Zero order Second order

✓ Check

Interactive 1. What order?

 More information for interactive 1

An interactive feature illustrates two graphs to help users understand reaction order. The first graph (on the left of the interface) plots concentration on the y-axis and time on the x-axis. The second graph (on the right of the interface) plots rate on the y-axis and concentration on the x-axis.

The first graph exhibits two concave upward decreasing curves, one decreasing more steeply than the other, and a straight line with a negative gradient.

The second graph features a straight line passing through the origin with a positive gradient, a concave upward increasing curve, and a horizontal, stable line in the center.

Users are required to determine the reaction order: zero, first, or second. They must then drag and drop the appropriate labels onto the corresponding curves in both graphs.

This interactivity helps users to distinguish between zero-, first-, and second-order reactions using concentration versus time and rate versus concentration graphs.

Read below for the solution:

First graph (on the left of the interface):

More steeply decreasing concave curve - second order

Less steeply decreasing concave curve - first order

A straight line with a negative gradient - zero order

Second graph (on the right of the interface):

A straight line passing through the origin with a positive gradient - first order

Concave upward increasing curve - second order

Horizontal, stable line - zero order

This activity helps users learn how to recognize different reaction orders by analyzing the shapes of concentration-time and rate-concentration graphs. It provides an interactive way to understand the relationship between reaction rate and concentration.

Orders of reaction and reaction mechanisms

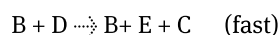
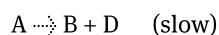
In [section R2.2.9 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-rate-expression-and-its-experimental-id-45511/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-rate-expression-and-its-experimental-id-45511/) you learned that the rate expression can only be determined experimentally. Although we know the molecularity of the overall reaction and can predict how many particles are colliding, why can we not predict their effect on the rate from the stoichiometry of the reaction?

It so happens that the rate expression cannot be deduced from the stoichiometry of the reaction. This only applies to overall reactions, and not elementary steps. Let's see how the order of reactions is related to the elementary steps in a reaction mechanism.

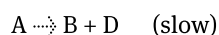
Cases where the slow step is the first step in the reaction mechanism

Overall reaction $2A \rightarrow 4B + C$

The order of this reaction was experimentally determined to be first order. The overall order will depend on the rate law from the rate determining step in the reaction mechanism. Since the order of the elementary steps can be determined from the reaction coefficients, we need to examine the proposed mechanism. The proposed mechanism for this reaction is:



Because the rate determining step is an elementary step, its rate law comes from its molecularity. Collision theory states that the concentration of each reactant in the rate determining step must appear in the rate law for that step, raised to the power of its stoichiometric coefficient in the equation. The slow step is:



The molecularity of the rate-determining step is one. That is why the rate law for this step is:

$$\text{rate} = k[A]$$

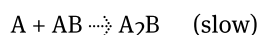
Table 1. The relationship between the molecularity of elementary steps and their rate law

Elementary step	Molecularity	Rate law from the elementary step
$A \rightarrow \text{products}$	unimolecular	$\text{rate} = k[A]$
$2A \rightarrow \text{products}$	bimolecular	$\text{rate} = k[A]^2$
$A + B \rightarrow \text{products}$	bimolecular	$\text{rate} = k[A][B]$

The rate law for the rate determining step leads us to the rate expression for the overall reaction. If the rate determining step is the first step in the reaction mechanism or the only step in a single step mechanism, then its rate expression is the same as the rate expression for the overall reaction.

Cases where the slow step is not the first step in the reaction mechanism

If the rate determining step is not the first step in the reaction mechanism, then deducing the reaction mechanism becomes a little more complicated. When the rate determining step comes after initial elementary steps the concentration of the other species in the previous steps need to be considered, as they also influence the rate. Consider a reaction between A and B:



The slow step is the second step. The rate of reaction will be governed by the slow step, so the rate expression will look like this:

$$\text{rate} = k[A][AB]$$

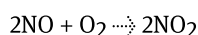
Except of course AB is not one of the reactants we started with. AB here is an intermediate. The rate expression is everything up to and including the slow step, except for any reaction intermediates. The concentration of AB must be expressed in terms of [A] and [B]. We can see that from the first equation that AB is formed from A + B. So substituting this into the equation instead of [AB] we get:

$$\begin{aligned}\text{rate} &= k[A][A][B] \\ &= [A]^2[B]\end{aligned}$$

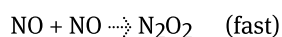
So the reaction is second order with respect to A and first order with respect to B.

Worked example 2

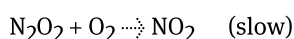
For the reaction:



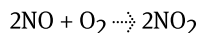
Step 1:



Step 2:



Step 2 is the rate determining step.

Overall:

Write the rate expression for the overall equation.

What is the order of the overall reaction?

The rate depends on step 2 for which the rate expression is:

$$\text{rate} = k [\text{N}_2\text{O}_2][\text{O}_2]$$

But N_2O_2 is not a reactant in the overall equation. It is a product of step 1 that is formed by $\text{NO} + \text{NO}$, that is, $[\text{NO}]^2$.

Substituting this into the rate equation gives:

$$\text{rate} = k [\text{NO}]^2[\text{O}_2]$$

Hence this is a third order reaction ($2 + 1 = 3$).

Aspect:

- Evidence
- Falsification

More than one reaction mechanism can be proposed for a chemical reaction. We cannot prove which mechanism is definitely correct for that reaction unless we can support it with evidence. On the other hand we can eliminate a mechanism if it is proven wrong or falsified. In the past where advancements in techniques like spectroscopy have led to the identification and quantification of a new intermediate, certain proposed mechanisms have been falsified.

Units of k vary depending on the order of reaction

Table 2 shows how the units of k can be calculated from the rate expression when the concentration of the reactants and the corresponding rate is known. It can be seen that the units of k depend on the reaction order.

Calculating units of k for a first order reaction we have:

$$\text{rate} = k[\text{A}]$$

$$k = \frac{\text{rate}}{\text{A}}$$

Evaluating the units gives:

$$\begin{aligned} k &= \frac{\text{mol dm}^{-3} \text{s}^{-1}}{\text{mol dm}^{-3}} \\ &= \text{s}^{-1} \end{aligned}$$

The units of k for zero order, second order and third order reaction can be calculated similarly as shown in **Table 2**.

Table 2. Calculating the units of k .

Zero order	First order	Second order
rate = k	rate = $k[A]$	rate = $k[A]^2$
$k = \text{rate}$	$k = \frac{\text{rate}}{[A]}$	$k = \frac{\text{rate}}{[A]^2}$
Substituting for units	Substituting for units	Substituting for u
$k = \text{mol dm}^{-3} \text{ s}^{-1}$	$k = \frac{\text{mol dm}^{-3} \text{ s}^{-1}}{\text{mol dm}^{-3}}$	$k = \frac{\text{mol dm}^{-3} \text{ s}}{(\text{mol dm}^{-3})^2}$
	$k = \text{s}^{-1}$	$k = \text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$

Worked example 3

A reaction involving A and B is found to be first order with respect to A and zero order with respect to B.

The rate of reaction is $2.50 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$, when the initial concentrations of [A] and [B] are both $0.800 \times 10^{-3} \text{ mol dm}^{-3}$.

Calculate the value of the rate constant including its units.

The rate expression for this reaction is rate = $k[A]$.

Substituting the values into the rate expression gives:

$$\begin{aligned}
 2.50 \times 10^{-3} &= k \times 0.800 \times 10^{-3} \\
 k &= \frac{2.50 \times 10^{-3}}{0.800 \times 10^{-3}} \\
 &= 3.125 \\
 &= 3.13 \text{ (3 s.f.)}
 \end{aligned}$$

Substituting the units into the rate expression gives:

$$\begin{aligned}
 \text{mol dm}^{-3} \text{ s}^{-1} &= k \times \text{mol dm}^{-3} \\
 k &= \frac{\text{mol dm}^{-3} \text{ s}^{-1}}{\text{mol dm}^{-3}} \\
 &= \text{s}^{-1}
 \end{aligned}$$

Hence the rate constant is 3.13 s^{-1} (3 s.f.).

Activity

- **IB learner profile attribute:**
 - Knowledgeable
 - Inquirer

- Thinkers
- Reflective
- **Approaches to learning:**
 - Thinking skills — Providing a reasoned argument to support a conclusion, Applying key ideas and facts in new contexts
 - Communication skills — Using terminologies, symbols and communication conventions consistently and correctly
 - Social skills — Working collaboratively to achieve a common goal.
- **Time required to complete activity:** 40 minutes
- **Activity type:** Pair and individual activity

Activity 1

In this activity you will run a simulation in pairs. This is a half-life simulation.

Half-life is the time taken for the original concentration of the reactant to fall to half its original value. Radioisotopes follow this natural decomposition process where the time taken for the concentration of the radioisotope to fall to half is constant each time it decomposes.

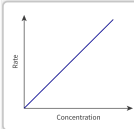
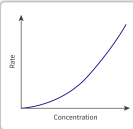
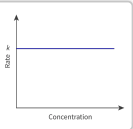
Open the simulation <https://academo.org/demos/radioactive-decay-simulator/>  (<https://academo.org/demos/radioactive-decay-simulator/>) and work through the following steps:

1. Start with 1600 atoms. Press the reset button. Enter a half-life of 20 in the space above. Enter 250 steps to be shown on the graph. Start the simulation by pressing the 'advance one step' button. Press the button 20 times to denote 20 steps.
 2. Record the number of atoms remaining after 20 steps. Round off to the nearest hundred.
 3. Next record the number of atoms remaining after another 20 steps. How are the number of atoms changing after each half-life of 20 steps?
 4. Continue pressing the button until the number of atoms remaining becomes zero. Describe the shape of the curve obtained. Which order of kinetics does this represent?
 5. Reset the simulation, this time enter a half-life of 40 steps. Tick the 'automatically advance steps' option and observe the graph formed. Describe the shape of this graph.
-
2. The number of atoms remaining after 20 steps is approximately 800.
 3. After a further 20 steps the total number of atoms remaining becomes approximately 400 hundred. After each half -life of 20 steps, the original number of atoms is halved.
 4. The graph is a curve with negative slope where the number of undecayed atoms halves after a fixed interval (half-life) of 20 steps, throughout the 250 steps. This represents first order kinetics.

5. The graph is similar in shape to D except it extends further as half-life is now increased.

Activity 2

Work individually to drag the cards under the right heading in **Interactive 2**.

Zero order	First order	Second order
Concentration–time graph		
Rate–concentration graph		
Rate = k		
Rate = $k[A]$	Rate = $k[A]^2$	Rate = k
		
		

☒ Check

Interactive 2. Features of orders of reaction.

 More information for interactive 2

An interactive table illustrates features of orders of reaction with missing fields. Users need to drag and drop the correct concentration-time graph, rate-concentration graph, and rate law expression ($\text{rate} = k$) for zero-order, first-order, and second-order reactions.

Drag and drop labels and graphs provided below the interface are;

$$\text{Rate} = k[A]$$

$$\text{Rate} = k[A]^2$$

$$\text{Rate} = k$$

A rate-concentration graph displays a straight line with a negative gradient.

A concentration-time graph displays a more steeply concave upward decreasing curve.

A rate-concentration graph displays a concave upward increasing curve.

A concentration-time graph displays a straight line passing through the origin with a positive gradient.

A concentration-time graph displays a less steeply concave upward decreasing curve.

A rate k-concentration graph displays a horizontal, stable line.

After completing the table, users can click the "Check" button to confirm whether their answers are correct and receive immediate feedback. This interactive table helps users to recognize how reaction order influences the shape of concentration-time and rate-concentration curves and apply rate laws.

Read below for the solution:

Zero order:

Concentration-time graph: Linearly decreasing curve

Rate-concentration graph: Horizontal, stable line

Rate law: $\text{Rate} = k$

First order:

Concentration-time graph: Less steeply concave upward decreasing curve

Rate-concentration graph: Straight line passing through the origin with a positive gradient.

Rate law: $\text{Rate} = k[A]$

Second order:

Concentration-time graph: More steeply concave with upward decreasing curve

Rate-concentration graph: Concave upward increasing curve

Rate law: $\text{Rate} = k[A]^2$

Arrhenius equation (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R2.2.12) Describe the qualitative relationship between temperature and the rate constant.
- (R2.2.12) Analyse graphical representations of the Arrhenius equation including its linear form.
- (R2.2.13) Determine the activation energy and Arrhenius factor from experimental data.

In section R2.2.9 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-rate-expression-and-its-experimental-id-45511/>) you learned that the rate of reaction depends on two things, the rate constant k and the concentration of the reactants raised to a power, represented by $\text{rate} = k [\text{A}][\text{B}]$.

You also learned about k in the previous section and how it is linked to the order of reaction and the rate of reaction. What exactly is k and what determines its value?

It is proportional to rate, so a higher value of k will mean a faster rate of reaction. Some reactions progress very quickly others are slow, is this because of k ?

The temperature dependence of the rate constant k

You have previously learned that increasing the temperature increases the rate of reaction (section R2.2.2–3a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/factors-affecting-the-rate-of-reaction-id-44714/>)). For example, milk will sour more quickly at higher temperatures than at lower ones. Since increasing temperature does not change the concentration of the reactants, it follows that the rate constant k is dependent on the temperature.

Generally speaking, the rate constant, k depends on two factors:

- The temperature — recall from the Maxwell-Boltzmann curves (section R2.2.4–5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/maxwell-boltzmann-distribution-curves-id-44782/>)) that at lower temperatures the number of collisions that have energy equal to or greater than activation energy are small. If you increase the temperature the number of collisions with energy equal to or greater than the activation energy grows exponentially. This means increasing the temperature will increase the rate constant k .
- The presence of a catalyst — this means an alternative route with lower activation energy. This increases the rate of reaction, hence increases k . The rate constant, k , is therefore also dependent on the value of activation energy.

It was Svante Arrhenius who consolidated these observations into a mathematical equation called the Arrhenius equation (**Table 1**).

Table 1.The Arrhenius equation.

Equation	Symbols	Units
$k = Ae^{-\frac{E_a}{RT}}$	k = rate constant	various units
	A = the Arrhenius factor	various units
	T = temperature	kelvin (K)
	R = gas constant (the value of which is given in the DP chemistry data booklet as $8.31 \text{ J K}^{-1}\text{mol}^{-1}$)	joules per mole per kelvin ($\text{J K}^{-1}\text{mol}^{-1}$)
	E_a = activation energy	joules per mole
	e = Euler's number This is the base of the natural logarithm (it has an approximate value of 2.718)	no units

The Arrhenius factor, A , is also called the pre-exponential factor. It takes into account the frequency and orientation of collisions. It varies slightly with temperature but not by much. It is generally considered constant across all temperatures and is specific for each reaction.

We know that for a successful collision to occur, particles must collide with:

- the proper orientation
- energy equal to or greater than the activation energy.

The Arrhenius factor takes into account the first point above. It tells us the frequency of collisions with the proper orientation, which is unique for every reaction.

It has the same units as k and can be deduced from the equation as follows:

$$A = \frac{k}{e^{-\frac{E_a}{RT}}}$$

The second point, the fraction of molecules with enough energy to react is represented by the next part of the equation that is:

$$e^{-\frac{E_a}{RT}}$$

This shows us that the fraction of molecules with enough energy depends on both temperature T , and the value of activation energy, E_a . The universal gas constant is a constant and does not impact the number of molecules with sufficient energy for a successful collision.

The Arrhenius equation can be rearranged mathematically, to determine the value of any of the variables in the equation depending on the given data:

- For any given reaction variables like temperature and rate (therefore rate constant) can be measured experimentally. R and e are constants.
- This leaves A and E_a , which must then be determined indirectly. Two unknowns require two equations, but as scientists we require several data points for validity, therefore graphing a series of data is required.
- The Arrhenius equation can be rearranged to form a linear equation. The independent and dependent variables can then be plotted as shown below. T is the independent variable as it is easily manipulated and k is the dependent variable, determined after rate is measured.

Graphical representation of the Arrhenius equation

1. Calculating the activation energy

Take the natural logarithm of both sides of the equation gives:

$$\begin{aligned}\ln k &= -\frac{E_a}{RT} + \ln A \\ &= -\left(\frac{E_a}{R}\right)\left(\frac{1}{T}\right) + \ln A\end{aligned}$$

This equation is in the form of a straight line equation $y = mx + c$.

Hence a graph of $\ln k$ (y-axis) against $\frac{1}{T}$ (x-axis) will be straight line with a gradient of $-\frac{E_a}{R}$.

This is known as the Arrhenius plot.

As R is a constant the activation energy can be calculated once the gradient of the graph is obtained:

$$\text{gradient} = -\frac{E_a}{R}$$

For example, if we were to measure the rate of degradation of a chemical at different temperatures, we could use these values to calculate the rate constant at those different temperatures from the rate expression $\text{rate} = k [\text{chemical}]$.

Plotting the $\ln k$ against $\frac{1}{T}$ will give the graph shown in **Figure 1**.

The gradient of the graph can be calculated and the value inserted into gradient $= -\frac{E_a}{R}$ to calculate E_a .

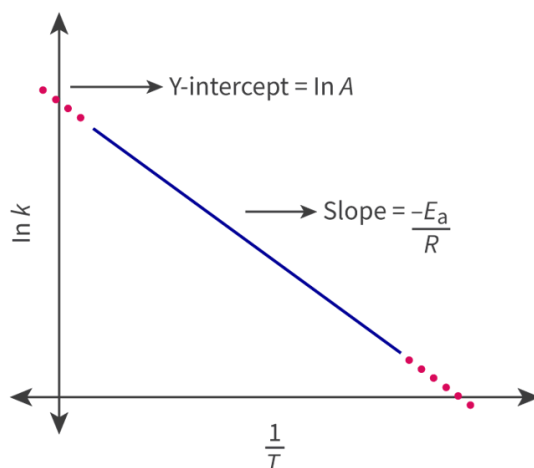


Figure 1. Arrhenius plot obtained by plotting experimentally obtained values of $\ln k$ vs $\frac{1}{T}$.

📌 More information for figure 1

2. Calculating the Arrhenius factor

Once we have drawn the Arrhenius plot using the experimental data we can determine the value of the Arrhenius factor as it is given by the y- intercept on the graph in the logarithmic form of the Arrhenius equation:

$$\ln k = -\frac{E_a}{RT} + \ln A$$

The graph can be extrapolated to find the value of $\ln A$, as $\ln A$ is the y-intercept as shown in **Figure 1**. This value can be used to calculate A .

Alternatively, the Arrhenius equation can be rearranged to:

$$A = \frac{k}{e^{-\frac{E_a}{RT}}}$$

and A can be calculated.

Study skills

- The temperature, T , is taken in kelvin in the Arrhenius equation. In order to convert degrees Celsius to kelvin just add 273.15. That is, $K = ^\circ\text{C} + 273.15$
- The activation energy, E_a , is given in J mol^{-1} . To convert to kJ mol^{-1} , divide by 1000.

Worked example 1

The following data was collected for a reaction.

Rate constant, k , (s^{-1})	Temperature ($^\circ\text{C}$)
0.00049	392

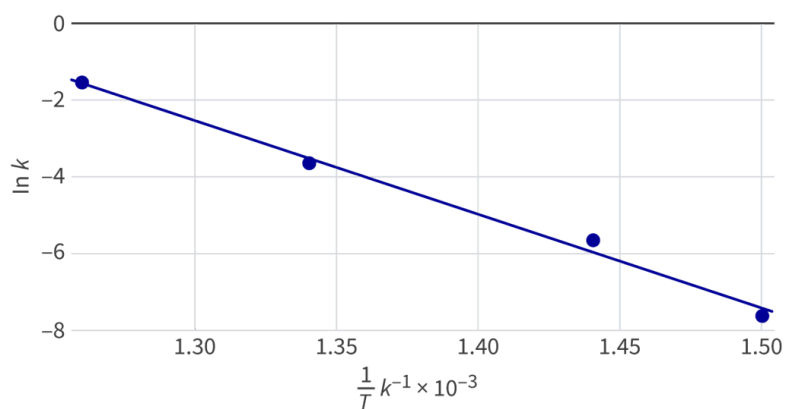
Rate constant, k , (s^{-1})	Temperature ($^{\circ}\text{C}$)
0.0035	421
0.026	475
0.21	521

1. Calculate the value of $\ln k$ and $\frac{1}{T}$
2. Plot the graph of $\ln k$ vs $\frac{1}{T}$
3. Measure the gradient of the line and calculate the activation energy in kJ mol^{-1} .

Convert the k values to $\ln k$ and the degrees Celsius to kelvin. Divide the kelvin values by 1 to get $\frac{1}{T}$.

$\ln k$	
-7.62	
-5.65	
-3.65	
-1.56	

Plotting the graph gives:



The gradient of the line is approximately -25000 K .

Using the equation;

$$\begin{aligned}\text{gradient} &= -\frac{E_a}{R} \\ -25000 &= -\frac{E_a}{8.31} \\ E_a &= (25000 \times 8.31) / 1000 \\ &= 208 \text{ kJ mol}^{-1}\end{aligned}$$

Please note that the final answer may change slightly depending on the value of the gradient.

Practical skills

Tool 1: Experimental techniques — Measuring variables, applying techniques

Tool 2: Technology — Applying technology to collect and manipulate data

Tool 3: Mathematics — Applying general mathematics and processing uncertainties

Inquiry 1: Exploring and designing — Controlling variables

Inquiry 2: Collecting and processing data — Interpreting results

Inquiry 3: Concluding and evaluating

A full investigation can be designed on determining the activation energy of a reaction using the Arrhenius equation, by varying the temperature of the reaction mixture.

Independent variable: Temperature

Dependent variable: k determined from the rate of reaction, calculated by measuring the change in concentration of the reactants over time at different temperatures.

Depending on the reaction under investigation, data loggers can be used to measure rate, followed by drawing an Arrhenius plot in Excel. Use the gradient of the plot to calculate E_a .

Calculation of activation energy without a plot

The activation energy can also be calculated using the values of the rate constant at only two different temperatures by using the equation:

$$\ln \left(\frac{k_1}{k_2} \right) = \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

Worked example 2

The following table shows data for the decomposition of hydrogen iodide at two different temperatures:

Rate Constant (k)	Temperature (T)
$k_1 = 7.78 \times 10^{-7}$	$T_1 = 273 \text{ K}$
$k_2 = 3.46 \times 10^{-5}$	$T_2 = 298 \text{ K}$

Calculate the activation energy for the reaction.

Rearranging and substituting the values into:

$$\ln \left(\frac{k_1}{k_2} \right) = \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

gives:

$$\begin{aligned}
 E_a &= \frac{R \ln \left(\frac{k_1}{k_2} \right)}{\left(\frac{1}{T_2} - \frac{1}{T_1} \right)} \\
 &= \frac{8.31 \ln \left(\frac{7.78 \times 10^{-7}}{3.46 \times 10^{-5}} \right)}{\left(\frac{1}{298} - \frac{1}{273} \right)} \\
 &= 1.03 \times 10^5 \text{ J mol}^{-1}
 \end{aligned}$$

Activity

- **IB learner profile attribute:**
 - Knowledgeable
 - Inquirer
 - Thinker
 - Reflective
- **Approaches to learning:**
 - Thinking skills — Providing a reasoned argument to support a conclusion, Applying key ideas and facts in new context
 - Communication skills — Using terminologies, symbols and communication conventions consistently and correctly
 - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 15–20 minutes
- **Activity type:** Pair activity

There is a familiar rule of thumb in chemistry which says that for reactions carried out at room temperature, for every 10 °C rise in temperature the rate of reaction doubles.

Perform the following activity in pairs and comment on the accuracy of this rule.

1. Use the Arrhenius equation to calculate the rate constants for a reaction where temperature is first 298 K and then it is increased to 308 K given the activation energy is taken as $E_a = 50 \text{ kJ mol}^{-1}$ at both temperatures. Assume that the Arrhenius factor is constant under all conditions.
2. Assuming the same 10 K rise in temperature, that is from 298 K to 308 K, calculate the new rate constants at 298 K and 308 K when the activation energy is 25 kJ mol^{-1} . Assume the Arrhenius factor is constant.
3. By what factor has the rate constant changed in both step 1 and step 2 (obtained by dividing the rate constant at 308 K with the rate constant at 298 K)?
4. Study the results obtained and brainstorm to comment on the reliability and accuracy of the above rule of thumb.

Doing a calculation and increasing temperature from to 298 K to 308 K and taking $E_a = 50 \text{ kJ mol}^{-1}$ ($48\,500 \text{ J mol}^{-1}$).

$$\begin{aligned} k_{298} &= Ae^{-\frac{E_a}{RT}} \\ &= Ae^{-\frac{50000}{8.31 \times 298}} \\ &= 1.70 \times 10^{-9} \text{ A} \end{aligned}$$

$$\begin{aligned} k_{308} &= Ae^{-\frac{E_a}{RT}} \\ &= Ae^{-\frac{50000}{8.31 \times 308}} \\ &= 3.28 \times 10^{-9} \text{ A} \end{aligned}$$

We can see that the rate constant has increased by a factor of $\frac{3.28 \times 10^{-9} \text{ A}}{1.70 \times 10^{-9} \text{ A}} = 1.9$ which is approximately twice the initial rate.

However, if we were to decrease the activation energy to 25 kJ mol^{-1} , substituting the value with the same rise in temperature gives:

$$\begin{aligned} k_{298} &= Ae^{-\frac{E_a}{RT}} \\ &= Ae^{-\frac{25000}{8.31 \times 298}} \\ &= 4.13 \times 10^{-5} \text{ A} \end{aligned}$$

$$\begin{aligned} k_{308} &= Ae^{-\frac{E_a}{RT}} \\ &= Ae^{-\frac{25000}{8.31 \times 308}} \\ &= 5.73 \times 10^{-5} \text{ A} \end{aligned}$$

For a 10 kelvin rise in temperature the rate constant increases by a factor of $\frac{5.73 \times 10^{-5} \text{ A}}{4.13 \times 10^{-5} \text{ A}} = 1.39$ which is approximately 1.4 times.

Activation energy values below and above 50 kJ mol^{-1} do not double the rate of reaction when temperature is increased by 10 K. The rule of thumb only applies to activation energies close to 50 kJ mol^{-1} and at temperatures close to room temperature.

The rule is not mathematically accurate.

